MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3130 **FOUNDATION DESIGN OF TRANSFER POINT TP-4**

DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1006

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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
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Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of TP-4
TP4 is the transfer point for conveyor 3130-FDR-003 and conveyor 3130-CNV-002

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of TP-4. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The pedestal reinforcement design has been performed in Staad-Pro Software. The footings and tie beams design have been performed in MS Excel.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

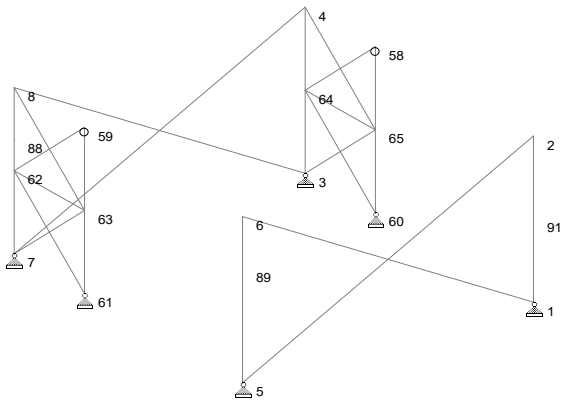
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3130-EG-ST-CAL-1006

5.1 Support Reaction at Grid 3130/A - 3130/6

		Horizontal	Vertical	Horizontal
Node	L/C	Fx kN	Fy kN	Fz kN
5	1 DEAD LOAD	1.5	22.4	0.0
	2 LIVE LOAD	3.9	34.6	0.0
	3 SDL1 - EQUIPMENT DEAD LO	12.4	82.1	0.0
	4 SDL2 - EQUIPMENT DEAD LO	9.0	81.8	0.0
	5 WL IN +X	-7.8	-14.1	0.0
	6 WL IN -X	6.3	7.9	0.0
	7 WL IN +Z	-0.6	3.4	-2.0
	8 WL IN -Z	1.3	-8.2	2.1
	9 TEMERATURE LOAD 1	66.3	-1.0	0.0
	10 TEMERATURE LOAD 2	-66.3	1.0	0.0
	11 EARTHQUAKE LOAD (+X)	-2.6	-3.0	0.0
	12 EARTHQUAKE LOAD (+Z)	-0.4	0.1	0.0



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5.2 Support Reaction at Grid 3130/A - 3130/8

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	-1.5	20.8	1.3
	2 LIVE LOAD	-3.8	48.0	4.3
	3 SDL1 - EQUIPMENT DEAD LO	-27.1	76.6	0.0
	4 SDL2 - EQUIPMENT DEAD LO	-21.0	73.9	0.0
	5 WL IN +X	-9.3	10.8	0.0
	6 WL IN -X	6.7	-7.5	0.0
	7 WL IN +Z	-0.1	1.2	-2.0
	8 WL IN -Z	0.9	-1.7	2.0
	9 TEMERATURE LOAD 1	-67.9	0.9	0.0
	10 TEMERATURE LOAD 2	67.9	-0.9	0.0
	11 EARTHQUAKE LOAD (+X)	-2.7	2.9	0.0
	12 EARTHQUAKE LOAD (+Z)	0.3	0.0	0.0

5.3 Support Reaction at Grid 3130/B - 3130/6

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
61	1 DEAD LOAD	0.0	9.4	-1.2
	2 LIVE LOAD	0.0	14.2	-1.8
	3 SDL1 - EQUIPMENT DEAD LO	0.0	-216.1	77.0
	4 SDL2 - EQUIPMENT DEAD LO	0.0	-125.6	45.9
	5 WL IN +X	-1.7	7.5	0.7
	6 WL IN -X	1.7	1.3	-0.8
	7 WL IN +Z	0.0	27.0	-11.4
	8 WL IN -Z	0.0	-50.9	20.3
	9 TEMERATURE LOAD 1	0.0	-2.5	12.1
	10 TEMERATURE LOAD 2	0.0	2.5	-12.1
	11 EARTHQUAKE LOAD (+X)	0.0	-1.1	0.5
	12 EARTHQUAKE LOAD (+Z)	0.0	7.2	-2.5
7	1 DEAD LOAD	1.7	25.9	1.2
	2 LIVE LOAD	3.1	37.2	1.7
	3 SDL1 - EQUIPMENT DEAD LO	10.7	310.5	69.5
	4 SDL2 - EQUIPMENT DEAD LO	8.4	219.6	43.5
	5 WL IN +X	-4.7	-11.4	1.3
	6 WL IN -X	3.8	3.4	-0.7
	7 WL IN +Z	-0.7	-30.4	-10.4
	8 WL IN -Z	0.5	59.1	18.3
	9 TEMERATURE LOAD 1	61.3	3.5	-10.3
	10 TEMERATURE LOAD 2	-61.3	-3.5	10.3
	11 EARTHQUAKE LOAD (+X)	-1.4	-0.4	0.4
	12 EARTHQUAKE LOAD (+Z)	-0.3	-7.2	-2.1

5.3 Support Reaction at Grid 3130/B - 3130/8

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
60	1 DEAD LOAD	0.0	10.8	-1.6
	2 LIVE LOAD	0.0	31.3	-4.5
	3 SDL1 - EQUIPMENT DEAD LO	0.0	-130.4	48.0
	4 SDL2 - EQUIPMENT DEAD LO	0.0	-71.7	27.9
	5 WL IN +X	-1.7	3.2	-1.2
	6 WL IN -X	1.7	-2.4	0.9
	7 WL IN +Z	0.0	15.1	-7.0
	8 WL IN -Z	0.0	-15.5	7.2
	9 TEMERATURE LOAD 1	0.0	2.8	8.8
	10 TEMERATURE LOAD 2	0.0	-2.8	-8.8
	11 EARTHQUAKE LOAD (+X)	0.0	1.4	-0.5
	12 EARTHQUAKE LOAD (+Z)	0.0	5.6	-1.9
3	1 DEAD LOAD	-1.7	16.8	0.4
	2 LIVE LOAD	-3.2	23.5	0.3
	3 SDL1 - EQUIPMENT DEAD LO	-22.4	217.5	44.6
	4 SDL2 - EQUIPMENT DEAD LO	-17.4	156.2	27.8
	5 WL IN +X	-5.6	3.9	-0.8
	6 WL IN -X	4.1	-2.6	0.6
	7 WL IN +Z	1.3	-16.2	-6.5
	8 WL IN -Z	-2.7	17.2	6.6
	9 TEMERATURE LOAD 1	-59.7	-3.7	-10.5
	10 TEMERATURE LOAD 2	59.7	3.7	10.5
	11 EARTHQUAKE LOAD (+X)	-1.5	0.2	-0.4
	12 EARTHQUAKE LOAD (+Z)	0.3	-5.6	-1.6

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5.5 LOAD COMBINATIONS

5.5.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD 301 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2)	
LOAD 303 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 304 (DL + SDL2 + TL1) + LL	
LOAD 305 (DL + SDL1 + TL2) + LL	
LOAD 306 (DL + SDL2 + TL2) + LL	
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X)	
LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z)	
LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z)	
LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X)	
LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X)	
LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z)	
LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X)	
LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z)	
LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z)	
LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X)	
LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X)	
LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z)	
LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	

5.5.2 Load Combination for Concrete design

Load Case	Reference
LOAD 401 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL + SDL2)	
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X)	
LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z)	
LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X)	
LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X)	
LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z)	
LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X)	
LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z)	
LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z)	
LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X)	
LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X)	
LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z)	
LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z)	
LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X)	
LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z)	
LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z)	
LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X)	
LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X)	
LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z)	
LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	

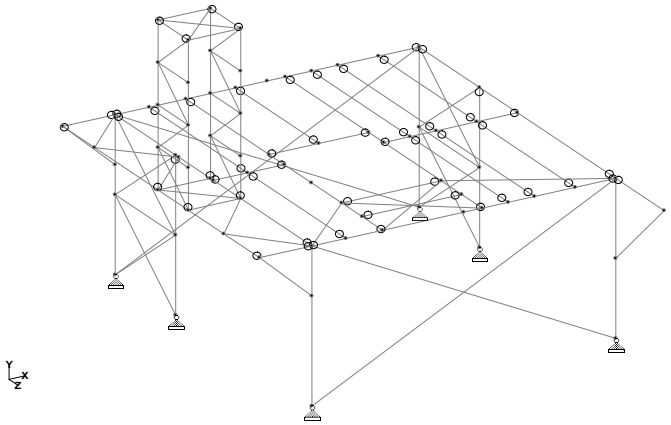
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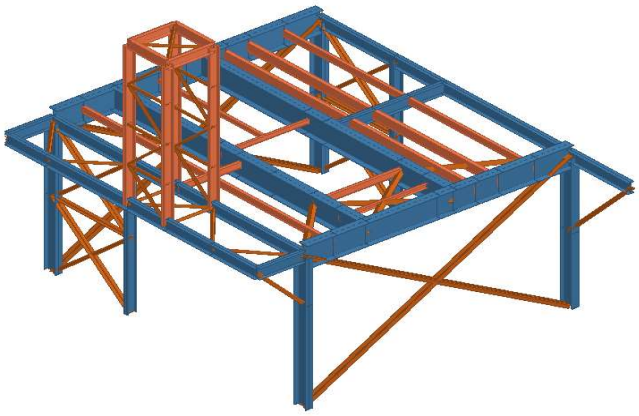
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6. GEOMETRY

6.2 Geometry of TP-4 Superstructure

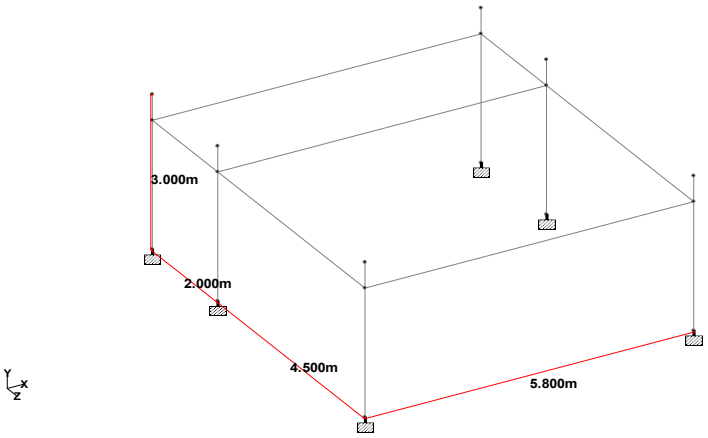


Staad Model of TP-4 Superstructure

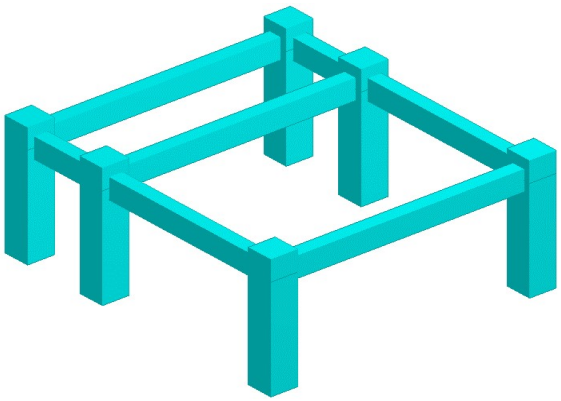


Isometric view of TP-4 Superstructure

6.3 Geometry of TP-4 Foundation



Staad Model of TP-4 Foundation framework



Isometric view of TP-4 Foundation framework

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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7. DESIGN OF CONCRETE COMPONENTS

All footings & tie beams have been designed using worksheet in accordance to SBC-304.

All columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachment 3 - Staad Output file.

7.1 Design Summary of Pedestal

7.1.1 Column Reinforcement Summary

Column mark	Dimension (mm)	Reinf. Area Required (mm^2)	Reinf. Area Provided (mm^2)	Reinforcement	Links	Remarks	Numbers
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/A-Grid 3130/6	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/A-Grid 3130/8	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/B-Grid 3130/6	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/B-Grid 3130/8	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/B-Grid 3130/6	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3130/B-Grid 3130/8	1

7.2 Design of Footing

7.2.1 Footing marking, Maximum Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (mm)	Width (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
Column at 3130/A - 3130/6	**F1**	**2500**	**2500**	**400**	**111**	**D12 @ 150c/c B/W**	**D12 @ 150c/c B/W**	**0.21**	**1**
Column at 3130/A - 3130/8	F1	2500	2500	400	100	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.19	1
Column at 3130/B - 3130/6	**F2**	**5500**	**3500**	**400**	**101**	**D12 @ 150c/c B/W**	**D16 @ 150c/c B/W**	**0.19**	**1**
Column at 3130/B - 3130/8	F2	5500	3500	400	84	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.16	1

The design of the most critical foundation of the group is attached in the next pages.

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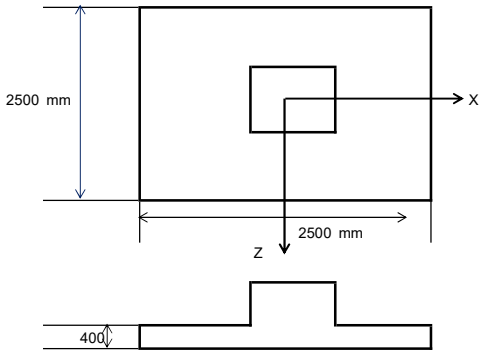
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7.2.2 Design of foundation F1 at Grid 3130/A-3130/6

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	169.63	KN
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	60.00	KN
Top of footing (staad support)	T.O.F	=	-1.600	m	Uplift	=	0.00	KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0	KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	36.7	KN/m^2
Length of footing	Lz	=	2.50	m				
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.40	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.400	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	12	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	6.3	132.77	-45.3	-105.91	-0.2	-16.2	362.4	104.87	11.10	1	0.20
302	4.0	137.95	-26.7	-63.86	-0.07	-12.1	367.6	87.97	29.66	1	0.16
303	21.9	165.8	-46.1	-107.60	-0.64	-29.7	395.4	116.01	10.53	1	0.22
304	19.6	170.98	-27.6	-65.54	-0.48	-25.6	400.6	99.11	29.08	1	0.18
305	-7.8	168.9	-44.6	-104.50	0.15	-3.7	398.5	105.32	22.21	1	0.20
306	-10.1	174.08	-26.0	-62.45	0.3	0.4	403.7	88.73	40.46	1	0.17
307	13.1	146.57	-45.7	-106.87	0.08	-33.1	376.2	113.92	6.46	1	0.21
308	2.1	165.86	-45.0	-105.27	-0.46	-4.0	395.5	105.25	21.30	1	0.20
309	7.0	162.75	-39.9	-93.19	-0.15	-17.2	392.4	105.17	20.39	1	0.20
310	6.0	150.03	-54.8	-128.06	-0.51	-14.4	379.7	115.46	6.03	1	0.22
311	10.9	151.76	-27.1	-64.80	0.2	-28.9	381.4	97.02	25.02	1	0.18
312	-0.2	171.04	-26.5	-63.21	-0.3	0.1	400.7	88.41	39.81	1	0.16
313	4.8	167.94	-21.3	-51.14	0.01	-13.1	397.6	88.27	38.95	1	0.16
314	3.8	155.22	-36.3	-86.00	-0.4	-10.3	384.9	98.57	24.58	1	0.18
315	8.1	156.77	-45.5	-106.33	-0.16	-20.1	386.4	110.36	13.29	1	0.21
316	5.5	160.65	-45.2	-105.70	-0.32	-13.1	390.3	108.07	16.82	1	0.20
317	6.8	158.96	-44.5	-104.26	-0.23	-16.6	388.6	108.57	15.78	1	0.20
318	6.9	158.45	-46.1	-107.78	-0.25	-16.6	388.1	109.85	14.33	1	0.20
319	5.9	161.95	-26.9	-64.28	0.0	-16.0	391.6	93.46	31.84	1	0.17
320	3.3	165.83	-26.6	-63.65	-0.16	-9.0	395.5	91.17	35.38	1	0.17
321	4.6	164.14	-26.0	-62.20	-0.07	-12.5	393.8	91.68	34.33	1	0.17
322	4.6	163.63	-27.6	-65.71	-0.09	-12.5	393.3	92.95	32.89	1	0.17
323	12.1	63.49	-27.6	-64.68	0.28	-31.7	293.1	83.90	9.90	1	0.16
324	-2.6	89.2	-26.7	-62.55	-0.4	7.0	318.8	77.73	24.30	1	0.15
325	4.0	85.06	-19.9	-46.45	-0.02	-10.5	314.7	72.23	28.47	1	0.13
326	2.7	68.1	-39.8	-92.94	-0.49	-6.8	297.7	85.95	9.32	1	0.16
327	10.8	66.6	-16.5	-39.44	0.38	-29.2	296.2	73.76	21.04	1	0.14
328	-3.9	92.31	-15.6	-37.32	-0.3	9.5	321.9	69.48	33.54	1	0.13
329	2.7	88.17	-8.8	-21.22	0.08	-8.1	317.8	62.09	39.61	1	0.12
330	1.3	71.21	-28.7	-67.70	-0.4	-4.4	300.8	75.81	20.46	1	0.14
331	5.5	77.07	-27.3	-63.97	-0.03	-14.3	306.7	79.14	19.00	1	0.15
332	2.0	82.25	-27.0	-63.13	-0.25	-5.1	311.9	76.09	23.71	1	0.14
333	3.7	80	-26.1	-61.21	-0.1	-9.7	309.6	76.77	22.31	1	0.14
334	3.8	79.32	-28.2	-65.89	-0.15	-9.7	309.0	78.47	20.40	1	0.15
335	4.2	80.18	-16.2	-38.74	0.07	-11.9	309.8	69.01	30.13	1	0.13
336	0.7	85.36	-15.9	-37.89	-0.15	-2.6	315.0	65.95	34.85	1	0.12
337	2.4	83.11	-15.0	-35.97	-0.03	-7.2	312.7	66.63	33.45	1	0.12
338	2.5	82.43	-17.1	-40.66	-0.1	-7.3	312.1	68.33	31.53	1	0.13

Max P_{max}	=	110.36	KN/m^2
Max P_{min}	=	6.03	KN/m^2
Resultant F.O.S	=	0.21	

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	8.8	185.9	-63.4	-148.28	-0.3	-22.66	95.38	-35.90
402	5.6	193.1	-37.4	-89.39	-0.1	-16.90	71.72	-9.92
403	30.8	242.5	-64.6	-150.68	-0.9	-41.81	112.71	-35.11
404	27.7	249.8	-38.6	-91.80	-0.67	-36.05	89.05	-9.13
405	-10.7	246.8	-62.4	-146.34	0.21	-5.34	97.74	-18.75
406	-13.9	254.1	-36.5	-87.46	0.44	0.41	74.40	6.91
407	21.3	150.7	-55.1	-128.97	0.39	-54.82	94.69	-46.46
408	-2.2	191.9	-53.7	-125.58	-0.75	7.08	81.64	-20.24
409	8.3	185.3	-42.8	-99.81	-0.09	-20.99	76.03	-16.75
410	6.2	158.1	-74.6	-174.19	-0.85	-15.10	97.98	-47.39
411	18.6	157.0	-32.9	-78.50	0.58	-49.89	74.42	-24.19
412	-4.9	198.1	-31.4	-75.11	-0.56	12.00	65.15	-1.75
413	5.6	191.5	-20.5	-49.33	0.1	-16.06	55.75	5.52
414	3.5	164.3	-52.4	-123.72	-0.66	-10.17	77.71	-25.12
415	10.4	172.9	-54.6	-127.76	-0.12	-26.32	86.83	-31.50
416	5.4	180.3	-54.1	-126.57	-0.44	-13.06	82.47	-24.77
417	7.8	177.1	-52.9	-123.82	-0.27	-19.66	83.43	-26.76
418	7.9	176.1	-55.8	-130.51	-0.29	-19.72	85.87	-29.51
419	7.7	179.1	-32.4	-77.30	0.07	-21.39	66.56	-9.23
420	2.7	186.5	-31.8	-76.10	-0.25	-8.13	62.19	-2.50
421	5.1	183.3	-30.6	-73.35	-0.07	-14.73	63.16	-4.49
422	5.3	182.3	-33.6	-80.04	-0.1	-14.79	65.59	-7.24
423	19.1	93.6	-41.5	-97.13	0.47	-49.70	71.36	-41.41
424	-4.5	134.8	-40.1	-93.73	-0.68	12.20	62.24	-19.12
425	6.1	128.1	-29.1	-67.97	-0.02	-15.86	52.69	-11.69
426	3.9	101.0	-61.0	-142.34	-0.78	-9.98	74.65	-42.33
427	17.0	98.3	-24.8	-59.28	0.61	-46.00	56.15	-24.70
428	-6.5	139.4	-23.4	-55.88	-0.53	15.90	49.87	-5.25
429	4.0	132.8	-12.5	-30.11	0.13	-12.17	37.48	5.01
430	1.9	105.7	-44.3	-104.49	-0.63	-6.28	59.44	-25.63
431	8.1	115.8	-41.0	-95.92	-0.05	-21.18	63.50	-26.44
432	3.1	123.2	-40.5	-94.73	-0.36	-7.94	59.14	-19.71
433	5.6	120.0	-39.3	-91.97	-0.19	-14.54	60.10	-21.70
434	5.7	119.0	-42.2	-98.66	-0.22	-14.60	62.53	-24.45
435	6.1	120.5	-24.3	-58.06	0.1	-17.49	48.29	-9.74
436	1.1	127.9	-23.8	-56.88	-0.22	-4.25	43.93	-3.02
437	3.6	124.7	-22.6	-54.12	-0.05	-10.84	44.89	-5.00
438	3.7	123.7	-25.5	-60.82	-0.07	-10.90	47.33	-7.75

Max Net factored B.P = 112.71 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: M_{uz} = 50.86 KNm

$$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

M_{uzlim} = 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

A_{st} = 425.3 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

$A_{stmin} = 0.0014bd$

A_{stmin} = 574.2 mm^2

A_{req} = 425.3 mm^2

Required Spacing = **197.0** mm

Length of cantilever along Z = 0.95 m

Moment about X: M_{ux} = 50.86 KNm

$$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

M_{uxlim} = 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

A_{st} = 442.2 mm^2

Effective Depth Along X = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

$A_{stmin} = 0.0014bd$

A_{stmin} = 552.6 mm^2

A_{req} = 442.2 mm^2

Required Spacing = **204.7** mm

Provide Bottom Reinforcement = **12 mm dia @ 150 c/c**

Ast prov = 754.0 mm^2

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Top Reinforcement

Length of cantilever along X = 0.95 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 287.1 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **393.9** mm

Length of cantilever along Z = 0.95 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 307 mm

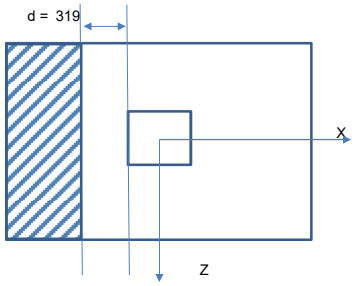
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 276.3 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **409.3** mm

Provide top Reinforcement = **12 mm dia @ 150 c/c**
Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.631 m
Reinforcement ratio: ρ_w = 0.0024
Effective Depth Along X: d = 319 mm
Shear Force at d distance: V_u = 71.12 KN
Allowable shear strength: φVc = 236.08 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'}b_w d + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*
λ = 1 Normal concrete
Mu = 22.44 KNm
φ = 0.75 for shear
φVc = 236.19 KN

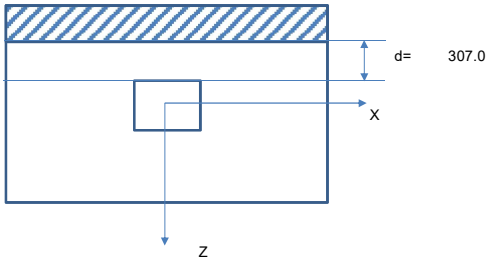
ii) $V_c = \left(0.16\lambda\sqrt{f_c'}b_w d + 17\rho_w\right)b_w d$
φVc = 236.08081 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 410.472 KN

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Project Country	**KINGDOM OF SAUDI ARABIA**

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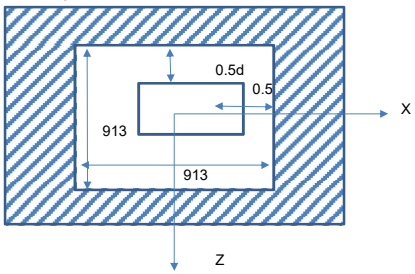
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.643	m	Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0025						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	307	mm			λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	72.48	KN			Mu	=	23.30 KNm
Allowable shear strength		φVc	=	227.13	KN	O.K		φ	=	0.75 for shear
								φVc	=	227.13 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	227.56165 KN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	395.03 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	6.25	m2	
Punching area	=	0.83	m2	
λ	=	1		
β	=	1		
αs	=	40		
b0	=	3652.0	mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	$= 0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$= 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$= 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

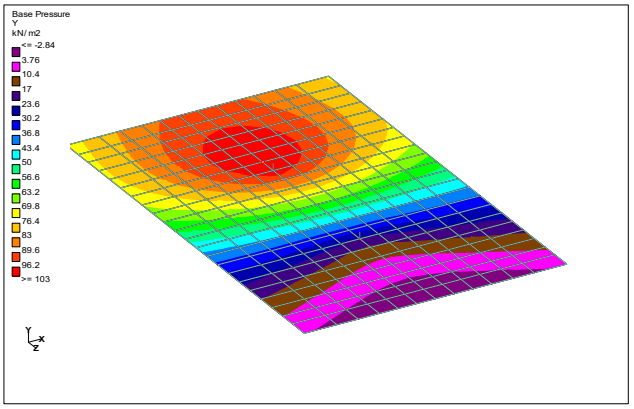
Perimeter at 0.5d from column face	u	=	3.65	m		Vc	= $v_c b_0 d$
Depth at section	d	=	313	mm			= 2231.63 KN
Shear Force at section	Vu	=	610.5	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	1673.73	KN	O.K	φV_n	= φVc
						φVc	= 1673.7 KN

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7.2.3 Design of combined footing mkd. as F2

For Calculation of Base pressure and design forces footing, Refer STAAD File **TP4-F2.STD**



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.500	m	
Width of footing	Lx	=	5.500	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	29	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	101.3	KN/m^2	Load Case 310
Minimum Base Pressure		=	-6.7	KN/m^2	Load Case 326

2) Design of foundation Raft :

Slab Thickness =	400	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	ength Reduction Factor of Shear Reinforcement , Φt =	0.75	

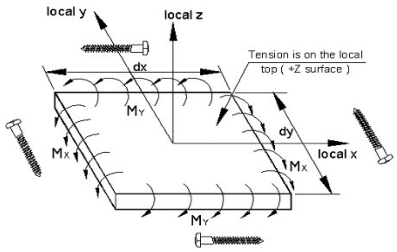
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	31.830	kN-m	Corresponding Mxy	1.380	kN-m	*(Plate No. - 24 , L/C No. - 410)*
Min. Mx	-65.250	kN-m	Corresponding Mxy	11.570	kN-m	*(Plate No. - 115 , L/C No. - 403)*
Max. My	85.990	kN-m	Corresponding Mxy	0.060	kN-m	*(Plate No. - 26 , L/C No. - 410)*
Min. My	-106.550	kN-m	Corresponding Mxy	4.870	kN-m	*(Plate No. - 114 , L/C No. - 410)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	31.83 + 1.38 =	33.210	kN-m
My1 =	My + abs(Mxy) =	85.99 + 0.06 =	86.050	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	31.83 + 0.022 =	31.852	kN-m
My2 =	My+abs(Mxy^2/Mx) =	85.99 + 0 =	85.99	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	33.21	kN-m
Myd =	86.05	kN-m

ALONG LOCAL X - DIRECTION :

Mu =	33.21	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	319	mm
Mu/bd^2 =	33.21X10^6/(1000X319^2) =	0.326	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0009	Reqd. Ast =	).0009X1000X319/100 =	2.87	cm^2		
Provided	12 mm @	150	C/C	(Provide Ast =	7.54	$)cm^2$	
				Total Provide Ast =	7.54	cm^2	O.K.
				Total Provide Pt =	0.0024		

ALONG LOCAL Y - DIRECTION :

Mu =	86.05	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	307	mm
Mu/bd^2 =	86.05X10^6/(1000X307^2) =	0.846	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0023	Reqd. Ast =	).0023X1000X319/100 =	7.06	cm^2		
Provided	12 mm @	150	C/C	(Provide Ast =	7.54	$)cm^2$	
				Total Provide Ast =	7.54	cm^2	O.K.
				Total Provide Pt =	0.0025		

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FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -65.25 - 11.57 = -76.82 kN-m
My1 = My - abs(Mxy) = -106.55 - 4.87 = -111.42 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -65.25 - 1.256 = -66.506 kN-m
My2 = My-abs(Mxy^2/Mx) = -106.55 - 0.363 = -106.187 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -76.82 kN-m
Myd = -111.42 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 76.82 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 76.82X10^6/(1000X317^2) = 0.764 N/mm^2 < deff. = 317 mm = 6.9 N/mm^2
Pt = 0.0021 Reqd. Ast = 0.0021X1000X317/100 = 6.66 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0042

ALONG LOCAL Y - DIRECTION :

Mu = 111.42 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 111.42X10^6/(1000X301^2) = 1.23 N/mm^2 < deff. = 301 mm = 6.9 N/mm^2
Pt = 0.0033 Reqd. Ast = 0.0033X1000X301/100 = 9.93 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0045

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.22 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 247 , L/C No. - 403)*
Vu = 69.74 kN

Provided Pt = 0.0042
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0042X69.74X10^3X317/76.82/10^6)X1000X317/10^3
= 306.577 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0042)X1000X317/10^3
= 322.697 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 306.577 kN
Φt*Vc = 0.75X306.577 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 229.933 kN **> 69.74T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.3 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 146 , L/C No. - 410)*
Vu = 90.3 kN

Pt provided = 0.0045
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0045X90.3X10^3X301/111.42/10^3)X1000X301/10^3
= 290.536 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0045)X1000X301/10^3
= 307.945 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 290.536 kN
Φt*Vc = 0.75X290.536 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 217.902 kN **> 90.3T , Hence O.K.**

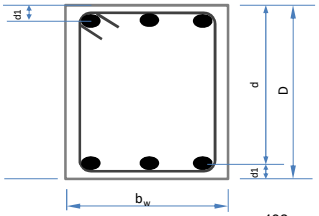
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7.3 Design of Tie Beams

7.3.1 Beam marked TB1

Staad beam number - 123,125,127,128, 133,134,135



Width of beam	=	**300**	mm				
Depth of beam	=	**500**	mm				
Grade of steel	=	420	MPa				
Grade of concrete	=	35	MPa				
Cover	=	50	mm				
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*			
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*			
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*			
Effective Depth	=	432	mm		d	=	432 mm
Dia of longitudinal bar	=	**16**	mm		d'	=	68 mm
Dia of stirrup bar	=	**10**	mm		h	=	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	0	Staad beam number -	0	=	**83**	KNm
B.M at span	Mz	L/C -	0	Staad beam number -	0	=	**71**	KNm
Shear Force	Fy	L/C -	0	Staad beam number -	0	=	**82**	KN

DESIGN FOR FLEXURE

Top Reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 83 KNm

Mu_2 = Mu - Mu_{lim} = 83 - 382.3

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 83 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 522.5 mm^2

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 456.4 mm^2 or 432 mm^2

= 456.4 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 522.6 + 0.00 + 0

= 522.6 mm^2

Top layer1		**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2			*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*
Ast provided	=	603.1857895	mm^2	Safe								
	ρ_w	=	0.00465421									

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Bottom reinforcement

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 71 KNm

Mu_2 = Mu - Mu_{lim} = 71 - 382.3

= 0 KNm

A_{st} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 71 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 445.0 mm²

= 0 mm²

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm²

= 456.4 mm² or 432 mm² = 456.3832976 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0 + 0

= 456.4 mm²

bottom layer1	3	nos	16	mm	dia	+		nos		mm	dia
bottom layer2		nos		mm	dia	+		nos		mm	dia

Ast provided = 603.1857895 mm² Safe

ρ_w = 0.004654211

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 127.1 KN λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 132.9 KN

$V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 222.3 KN

fV_c = 0.75 x 127.1 = 95.29 KN > 82 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((82-95.3) x 10^3)/(0.75 x 420 x 432) = 0 mm²/mm

No. of legs = 2 Spacing provided = 200 mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

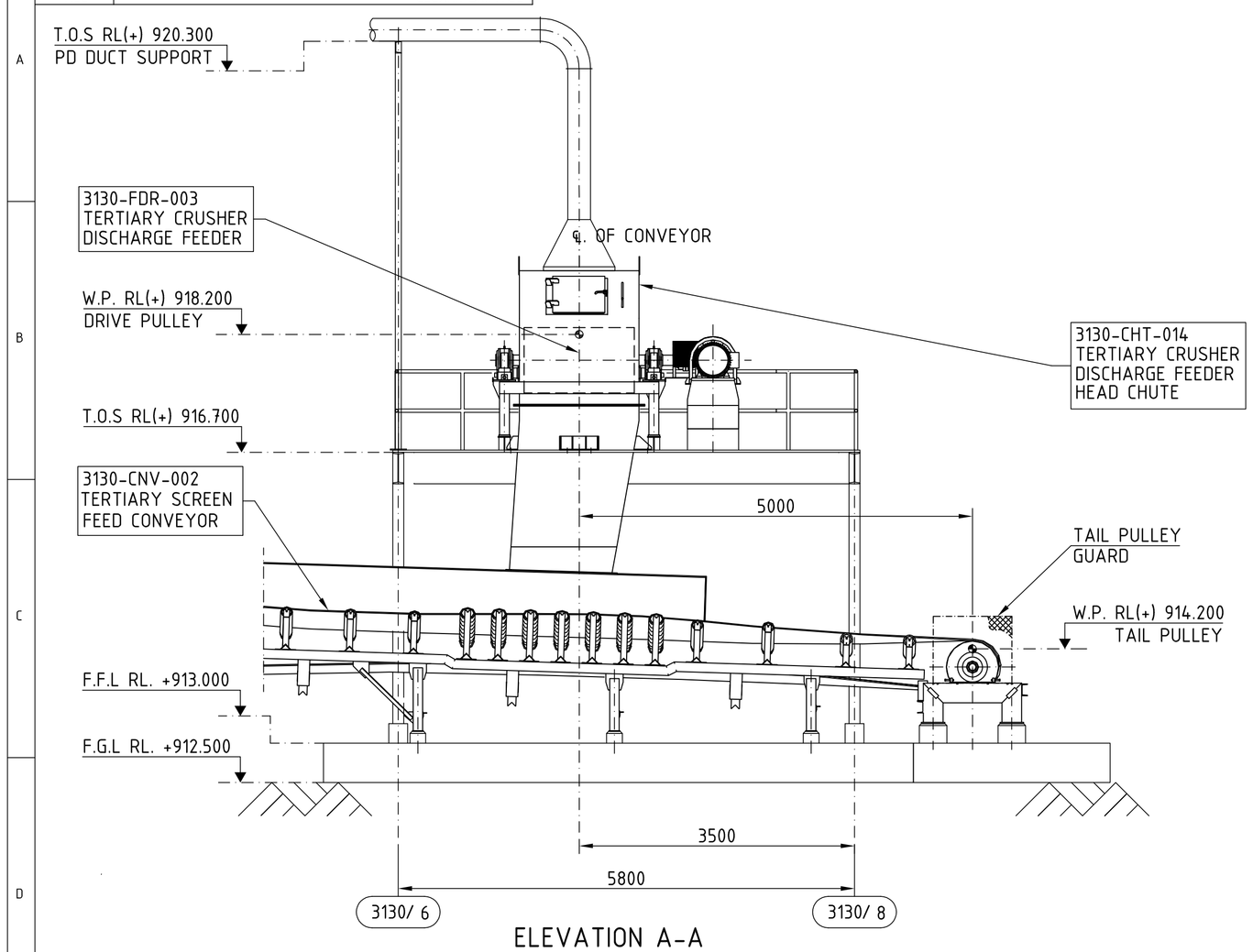
i) Shear Reinforcement	s =	0 mm	*if Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	216 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

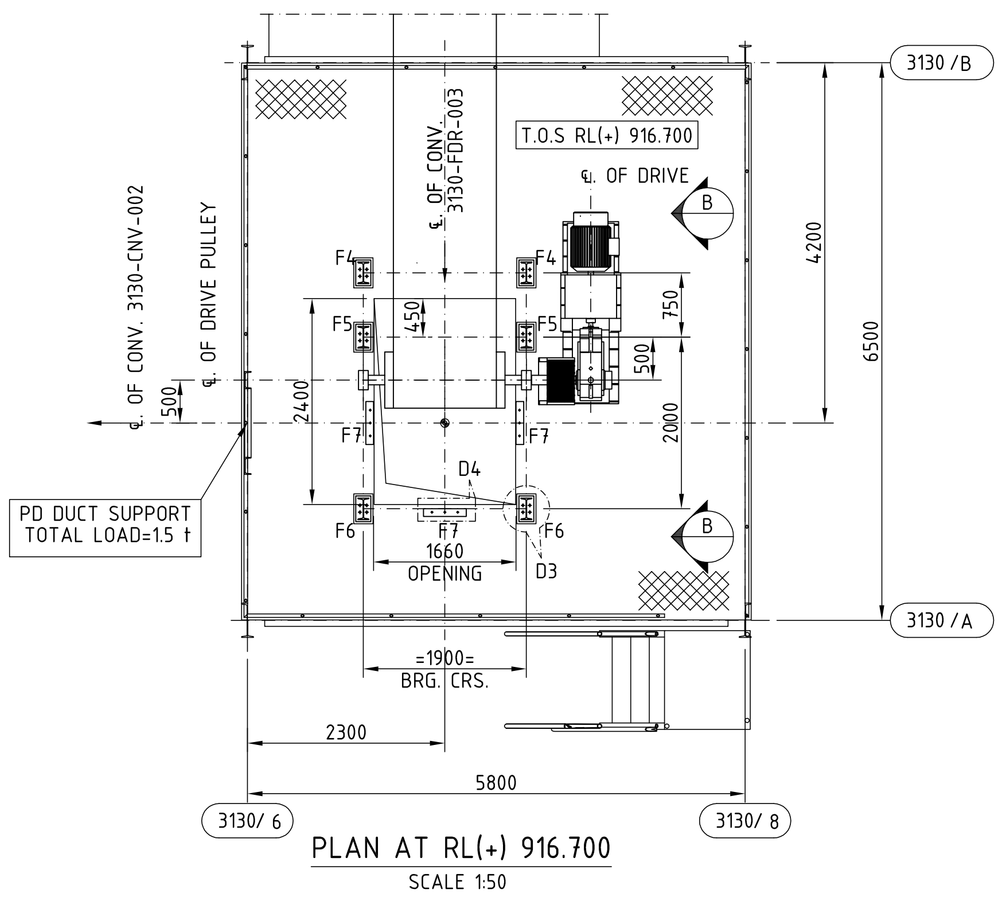
Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWINGS



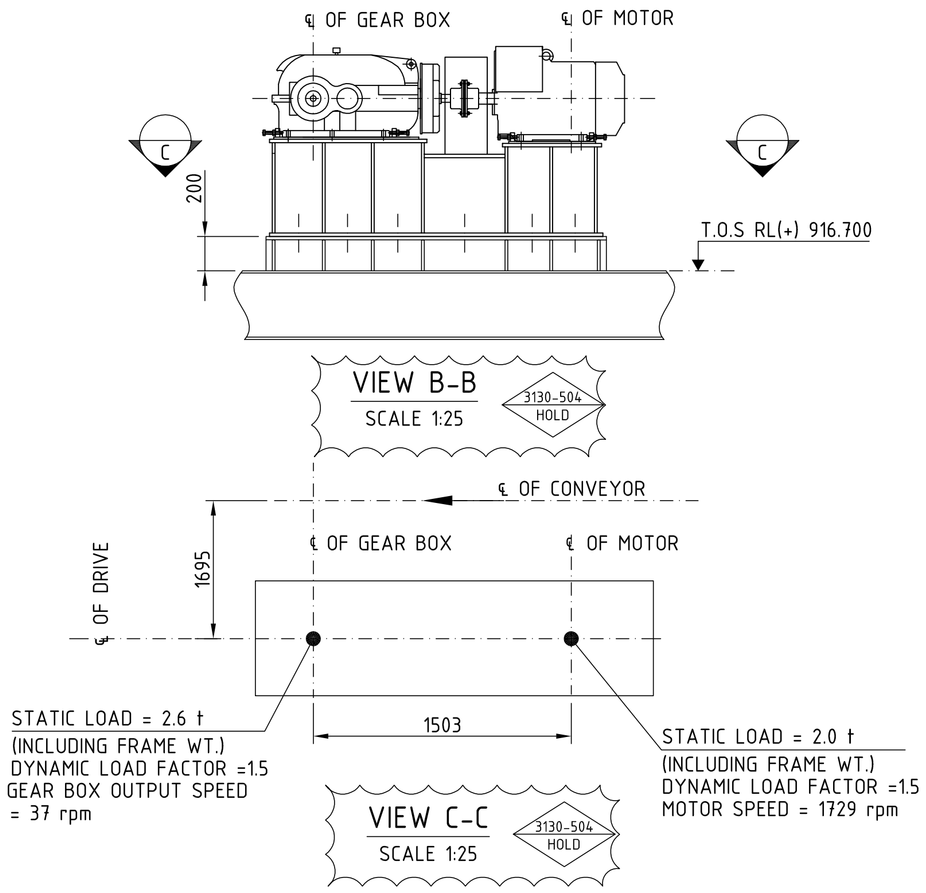
ELEVATION A-A

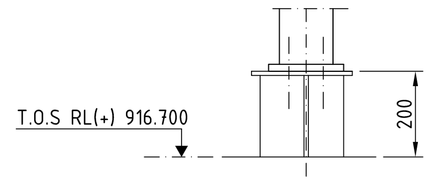
SCALE 1:50



PLAN AT RL(+) 916.700

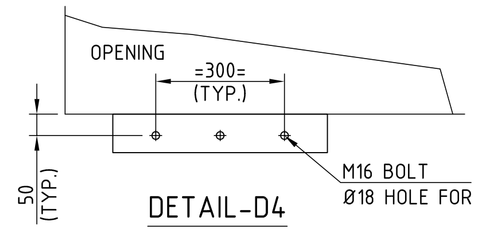
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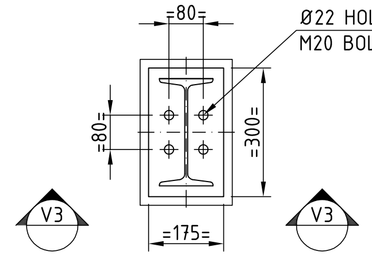
VIEW V3-V3

SCALE 1:10



DETAIL-D4

(TYPICAL)
SCALE 1:10



DETAIL-D3

SCALE 1:10

ABBREVIATION	
CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

TABLE-2 : FOR DRIVE PULLEY FRAME LOAD

FOR CONV. 3130-FDR-003

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F4	(+)916.700	(+) 5.40	(-) 3.80	(±) 0.55	(+) 8.51	(-) 6.00	(±) 0.85
F5		(+) 2.95	(-) 2.35	(±) 0.30	(+) 1.10	(-) 2.65	(±) 0.20
F6		(+) 1.41	(-) 1.10	(±) 0.20	(-) 2.65	(-) 3.30	(±) 0.27

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

TABLE-3: FOR CHUTE SADDLE LOAD

POINTS	ELEVATIONS IN "m"	V (IN "t")
F7	(+)916.700	(+)1.5

SIGN CONVENTION

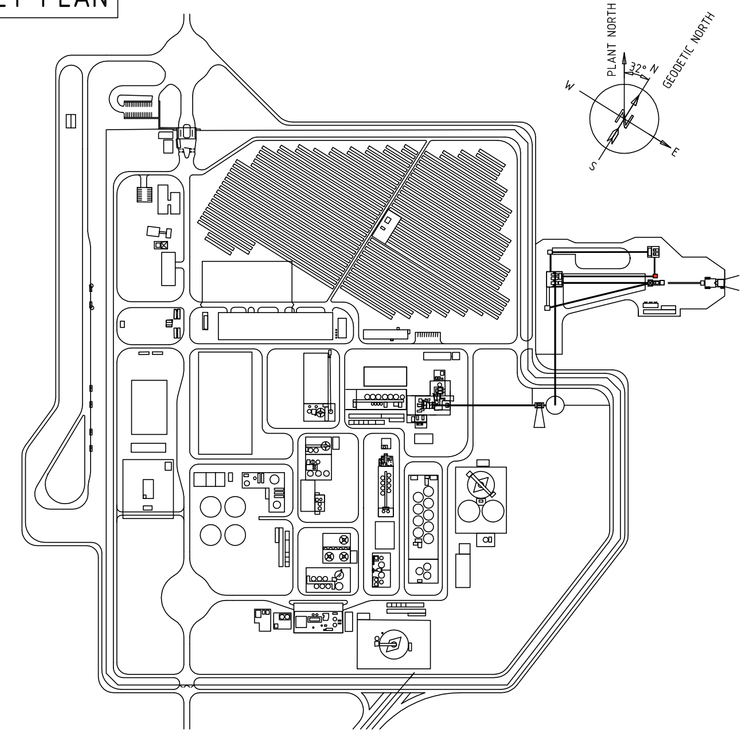
(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	01/11/2023	AS BUILT	SM	BRD	BD	BGR	
C01	19/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
		COMMENTS DT. 21.02.2020 & REVISED AS MARKED (1)					
A01	28/01/2020	ISSUED FOR APPROVAL	SM	BRD	BD	BGR	

DOCUMENT NUMBER: MD-216-3130-EG-ME-LPX-1016
REFERENCE DOCUMENTS: TRANSFER POINT TP#4 - PLAN VIEW

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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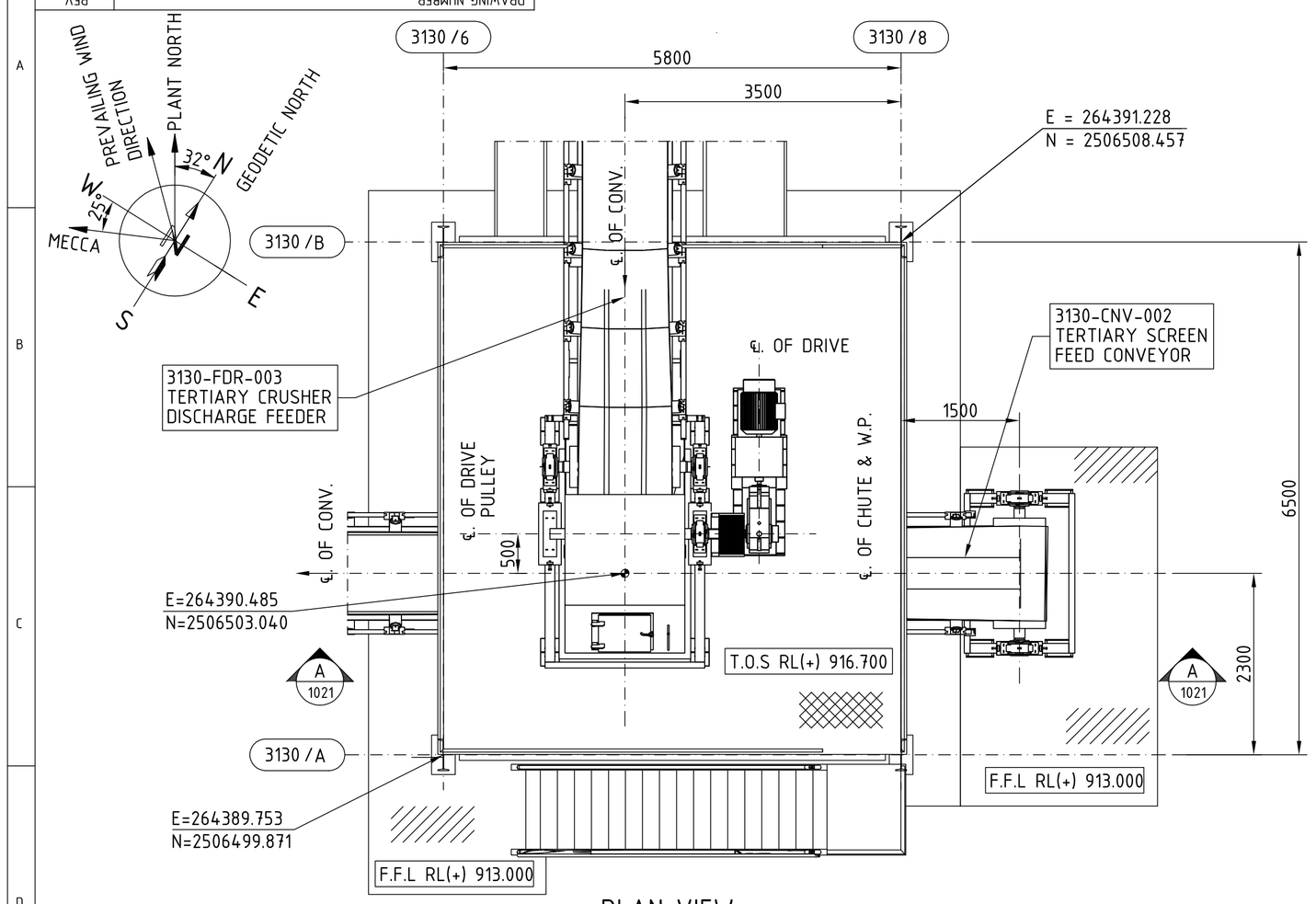
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16133_002 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31381 — DOCUMENT ID: OU100671070 — REVISION: 2

DRAWN	SM	21/01/2020
CHECKED	BRD	22/01/2020
ENGINEER	BD	24/01/2020
APPROVED	BGR	28/01/2020
EMPLOYER		
SCALE	1:50, 1:25 & 1;10	A1

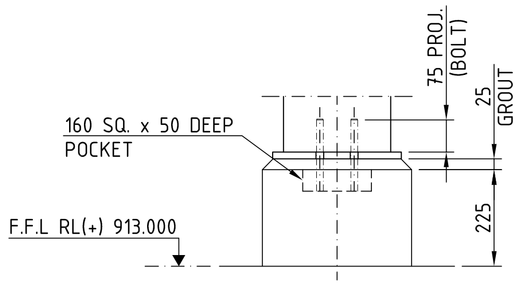
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TRANSFER POINT TP#4 - ELEVATION

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1021 — REV: D01

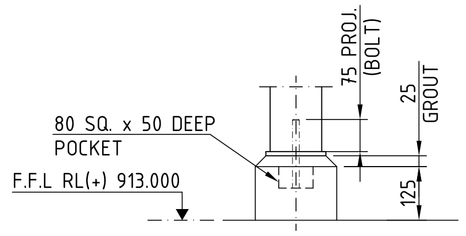
DRAWING NUMBER MD-216-3130-EG-ME-LPX-1016 REV D01



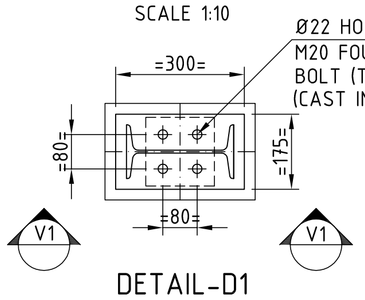
PLAN VIEW
SCALE 1:50



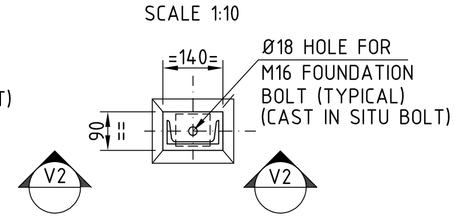
VIEW V1-V1
SCALE 1:10



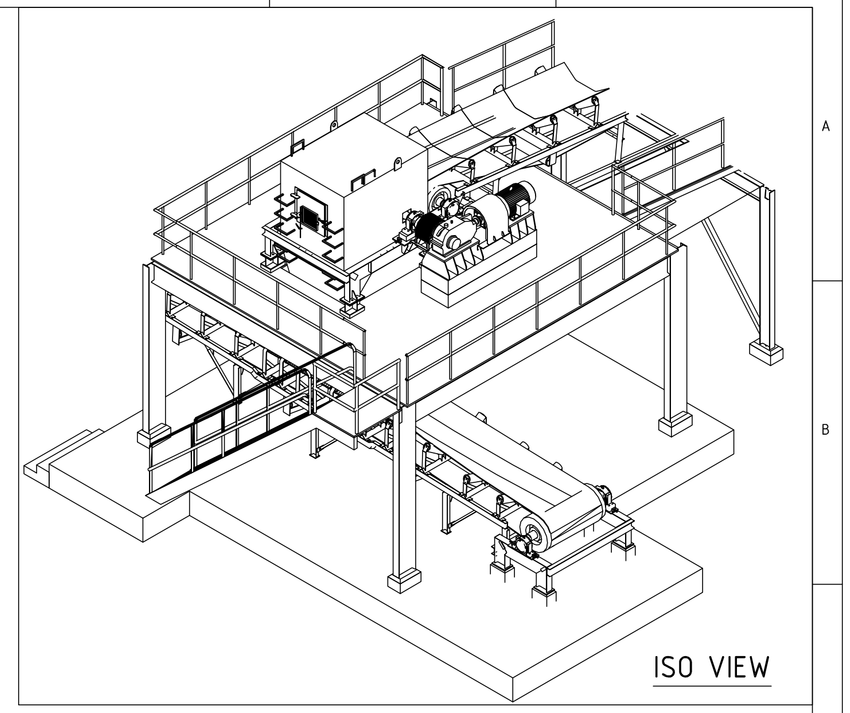
VIEW V2-V2
SCALE 1:10



DETAIL-D1
SCALE 1:10



DETAIL-D2
SCALE 1:10

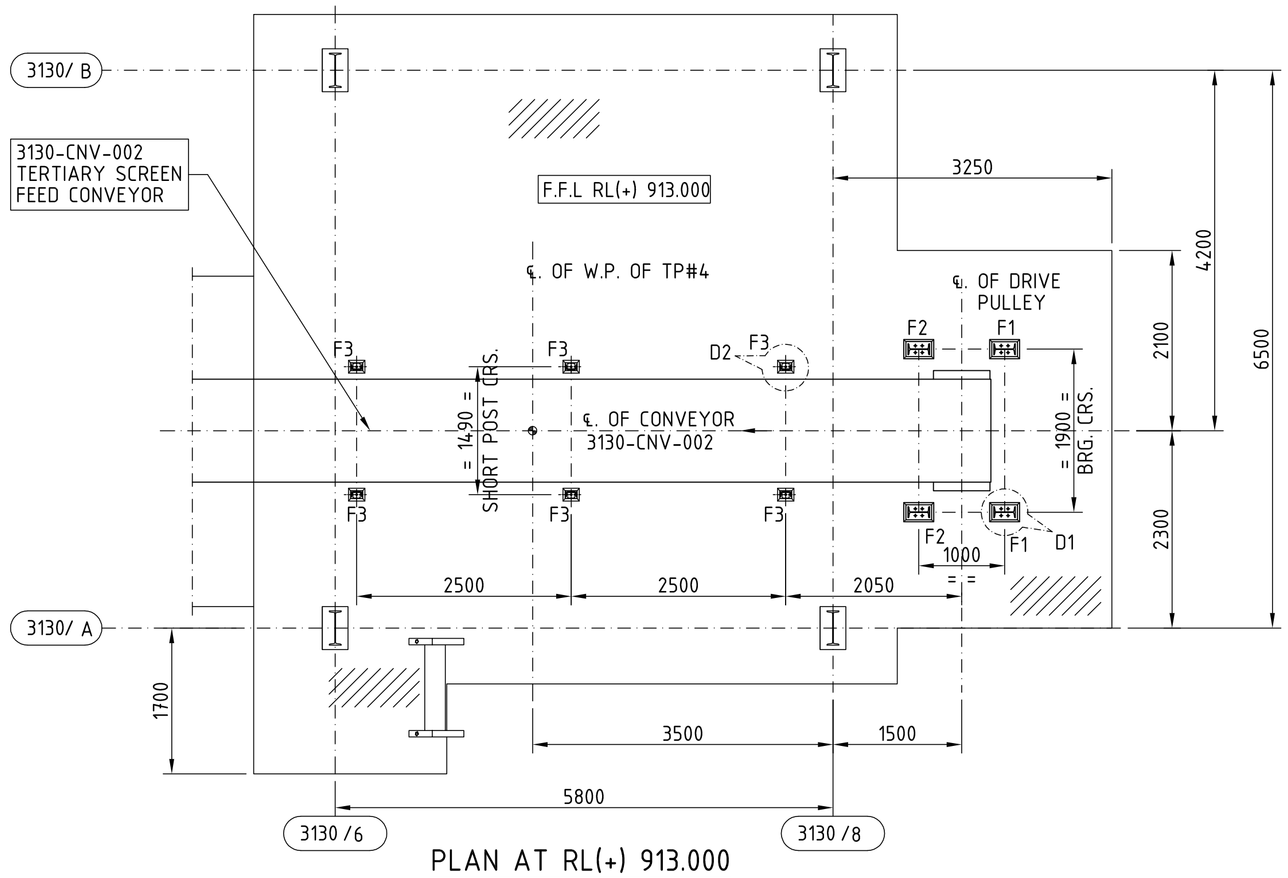


ISO VIEW

ABBREVIATION	
CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL



PLAN AT RL(+) 913.000
SCALE 1:50

TABLE-1 : FOR TAIL PULLEY FRAME LOAD

FOR CONV. 3130-CNV-002

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN t)	
			ALONG THE CONV. (IN "t")	ACROSS THE CONV. (IN "t")
F1	(+)913.000	(-)2.90	(+)2.40	(±)0.2
F2		(+)4.00	(+)2.90	(±)0.2

TABLE-2 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3130-CNV-002

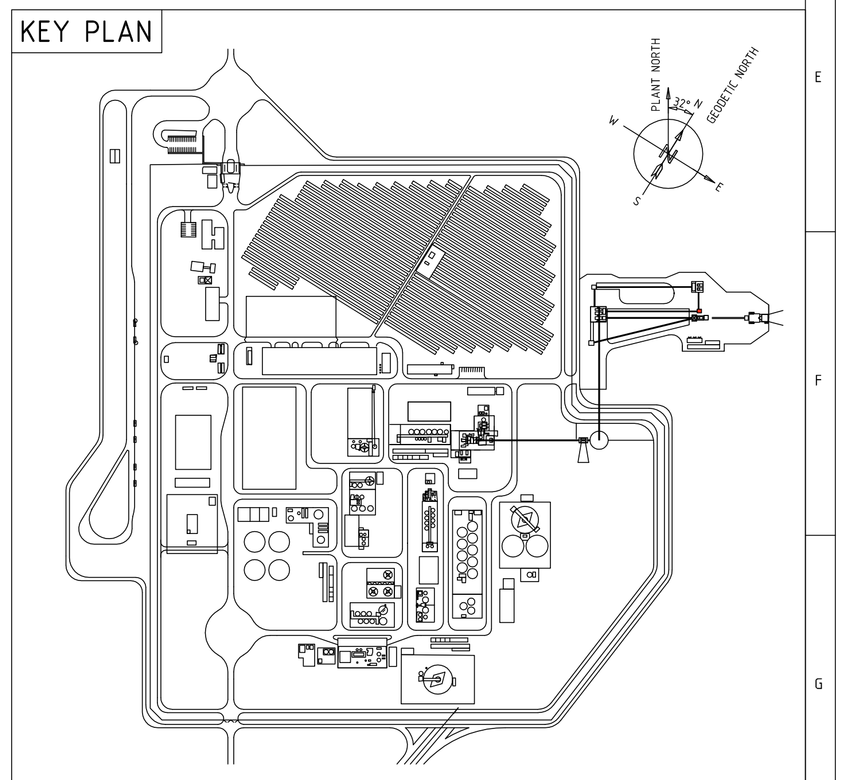
	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F3	(+)913.000	(+) 0.65	(+) 0.35	(±) 0.11	(±) 0.06

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	01/11/2023	AS BUILT	SM	BRD	BD	BGR	
		C01	19/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER COMMENTS DT. 21.02.2020 & REVISED AS MARKED	SM	BRD	BD	BGR	
MD-216-3130-EG-ME-ELV-1021	TRANSFER POINT TP#4 - ELEVATION	A01	28/01/2020	ISSUED FOR APPROVAL	SM	BRD	BD	BGR	

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DRAWN	SM	21/01/2020
CHECKED	BRD	22/01/2020
ENGINEER	BD	24/01/2020
APPROVED	BGR	28/01/2020
EMPLOYER		
SCALE	1:50, 1:10	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16133_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZC01 _MTL01 _31380 — DOCUMENT ID: 0U100671069 — REVISION: 2

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TRANSFER POINT TP#4 - PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1016 — REV: D01

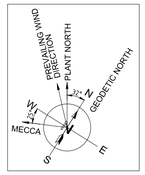
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

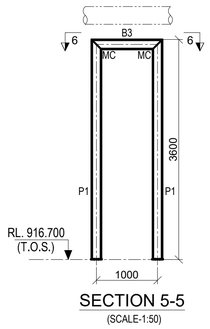
Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

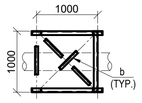
ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE LAYOUT**

DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1026 | REV: A01

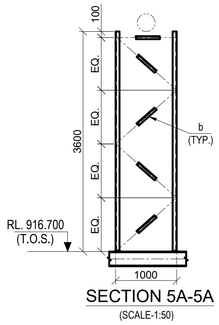




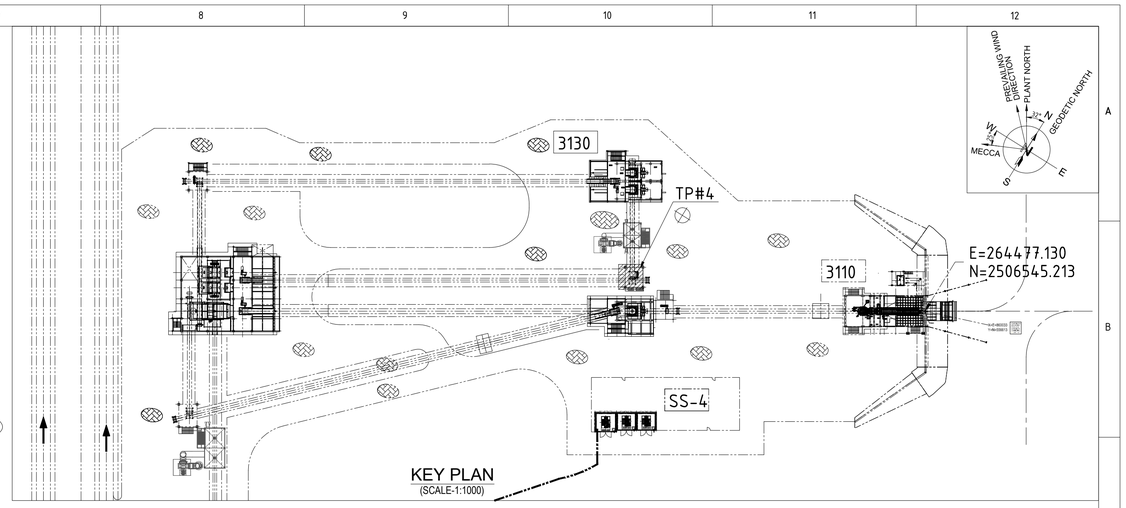
SECTION 5-5
(SCALE-1:50)



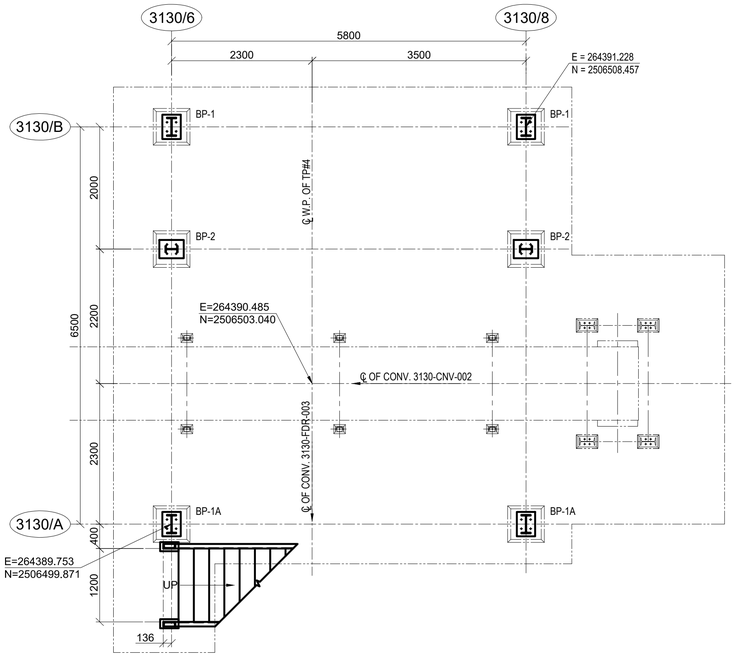
PLAN 6-6
(SCALE-1:50)



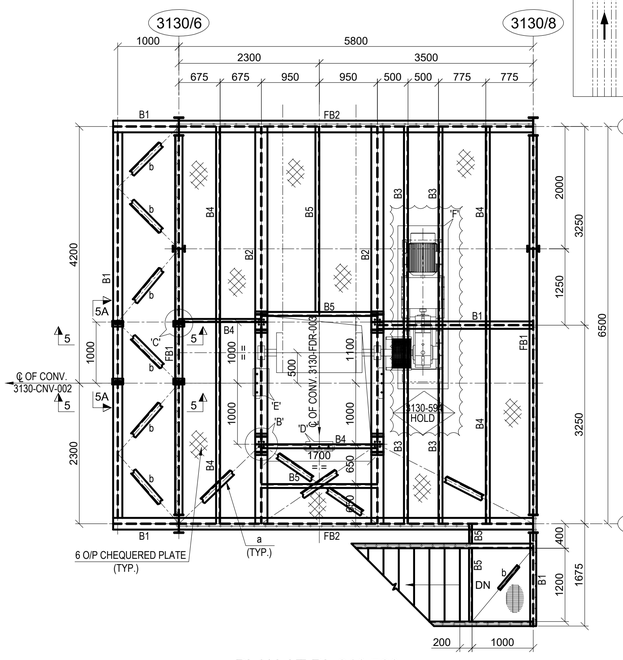
SECTION 5A-5A
(SCALE-1:50)



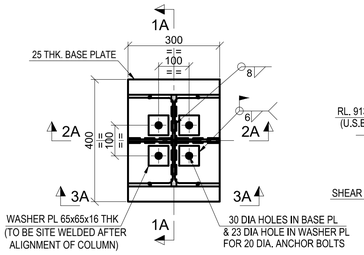
KEY PLAN
(SCALE-1:1000)



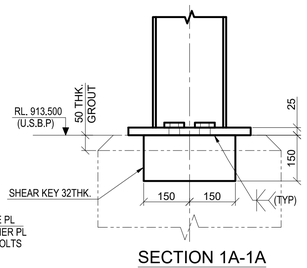
PLAN AT BASE LEVEL
(U/S OF BASE PLATE AT RL. 913.500 UNLESS NOTED OTHER WISE)
(SCALE - 1:50)



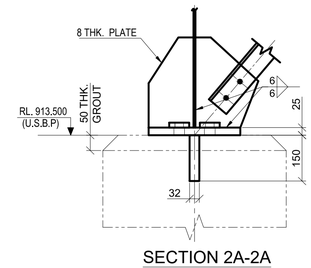
PLAN AT RL.916.700
(SCALE - 1:50)
(TOP OF FLOOR BRACING SHOULD BE KEPT 175mm BELOW T.O.S.)



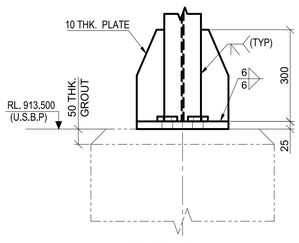
DETAIL OF BASE PLATE MKD. BP-1A
(SCALE - 1:10)



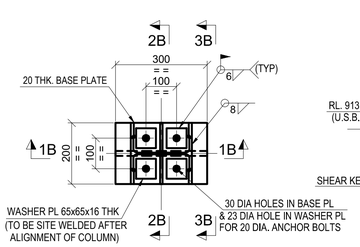
SECTION 1A-1A
(SCALE - 1:10)



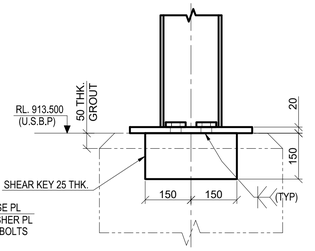
SECTION 2A-2A
(SCALE - 1:10)



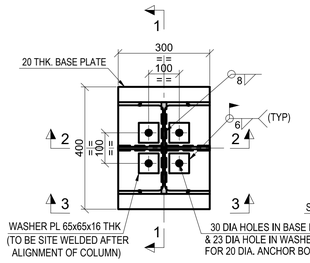
SECTION 3A-3A
(SCALE - 1:10)



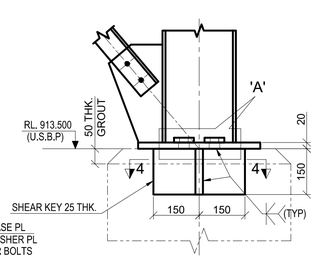
DETAIL OF BASE PLATE MKD. BP2
(SCALE - 1:10)



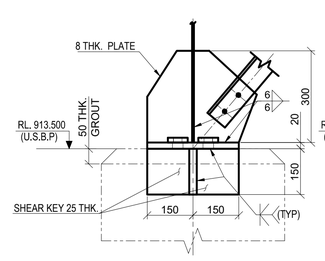
SECTION 1B-1B
(SCALE - 1:10)



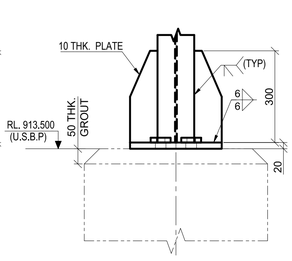
DETAIL OF BASE PLATE MKD. BP-1
(SCALE - 1:10)



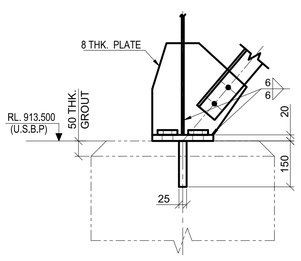
SECTION 1-1
(SCALE - 1:10)



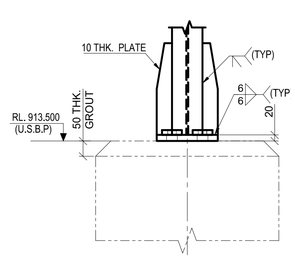
SECTION 2-2
(SCALE - 1:10)



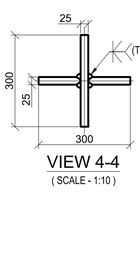
SECTION 3-3
(SCALE - 1:10)



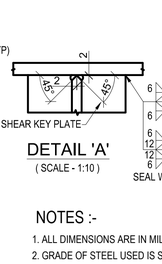
SECTION 2B-2B
(SCALE - 1:10)



SECTION 3B-3B
(SCALE - 1:10)



VIEW 4-4
(SCALE - 1:10)



DETAIL 'A'
(SCALE - 1:10)

LEGEND :-
- R.L. - REDUCED LEVEL
- ℄ - CENTRE LINE
- T.O.S. - TOP OF STEEL.
- F.G.L. - FINISHED GROUND LEVEL.
- F.F.L. - FINISHED FLOOR LEVEL.
- TYP. - TYPICAL.
- MKD. - MARKED.
- THK. - THICK
- W.P.-WORKING POINT
- U.S.B.P. - UNDER SIDE OF BASE PLATE
- U.N.O. - UNLESS NOTED OTHERWISE

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1027
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1016	TRANSFER POINT TP#4 - PLAN VIEW								
MD-216-3130-EG-ME-ELEV-1021	TRANSFER POINT TP#4 - ELEVATION	A01	07/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_S_G_26205_003		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31313	DOCUMENT ID 0U100668074	REVISION 0

DRAWN	SUS	05/04/2020
CHECKED	SBY	06/04/2020
ENGINEER	APR	05/04/2020
APPROVED	PRI	07/04/2020
EMPLOYER		
SCALE	1:100,20,15	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA OF TP-4(3130 CNV-001)-1
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1026 | REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF TP-3 FOUNDATION**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 09-Apr-2020
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 5.8 1 4.5; 3 5.8 1 -2; 5 0 1 4.5; 7 0 1 -2; 60 5.8 1 0; 61 0 1 0; 104 0 -2 0;
105 0 -2 4.5; 106 5.8 -2 0; 107 5.8 -2 4.5; 108 0 0.5 0; 109 0 0.5 4.5;
110 5.8 0.5 4.5; 111 5.8 0.5 0; 112 0 -2 -2; 113 0 0.5 -2; 114 5.8 -2 -2;
115 5.8 0.5 -2;
MEMBER INCIDENCES
113 104 108; 114 105 109; 115 107 110; 116 106 111; 121 108 61; 122 109 5;
123 108 109; 124 110 1; 125 109 110; 126 111 60; 127 108 111; 128 111 110;
129 112 113; 130 114 115; 131 113 7; 132 115 3; 133 113 115; 134 113 108;
135 115 111;
START GROUP DEFINITION
FLOOR
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
ISOTROPIC ALUMINUM
E 6.89476e+007
POISSON 0.33
DENSITY 26.6018
ALPHA 2.3e-005
DAMP 0.03
END DEFINE MATERIAL
MEMBER PROPERTY FRENCH
113 TO 116 121 122 124 126 129 TO 132 PRIS YD 0.6 ZD 0.6
123 125 127 128 133 TO 135 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
104 TO 107 112 114 FIXED
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
JOINT LOAD
3 FX 1.7 FY -16.8 FZ -0.4
```

```
7 FX -1.7 FY -25.9 FZ -1.2
60 FX 0 FY -10.8 FZ 1.6
61 FX 0 FY -9.4 FZ 1.2
1 FX 1.5 FY -20.8 FZ -1.3
5 FX -1.5 FY -22.4 FZ 0
LOAD 2 LIVE LOAD
JOINT LOAD
3 FX 3.2 FY -23.5 FZ -0.3
7 FX -3.1 FY -37.2 FZ -1.7
60 FX 0 FY -31.3 FZ 4.5
61 FX 0 FY -14.2 FZ 1.8
1 FX 3.8 FY -48 FZ -4.3
5 FX -3.9 FY -34.6 FZ 0
LOAD 3 SDL-1    EQUIPMENT DEAD LOAD CASE -1
JOINT LOAD
3 FX 22.4 FY -217.5 FZ -44.6
7 FX -10.7 FY -310.5 FZ -69.5
60 FX 0 FY 130.4 FZ -48
61 FX 0 FY 216.1 FZ -77
1 FX 27.1 FY -76.6 FZ 0
5 FX -12.4 FY -82.1 FZ 0
LOAD 4 SDL-2    EQUIPMENT DEAD LOAD CASE -2
JOINT LOAD
3 FX 17.4 FY -156.2 FZ -27.8
7 FX -8.4 FY -219.6 FZ -43.5
60 FX 0 FY 71.7 FZ -27.9
61 FX 0 FY 125.6 FZ -45.9
1 FX 21 FY -73.9 FZ 0
5 FX -9 FY -81.8 FZ 0
LOAD 5 WIND LOAD (+X)
JOINT LOAD
3 FX 5.55 FY -3.93 FZ 0.8
7 FX 4.68 FY 11.35 FZ -1.31
60 FX 1.71 FY -3.21 FZ 1.2
61 FX 1.69 FY -7.51 FZ -0.73
1 FX 9.32 FY -10.76 FZ 0
5 FX 7.75 FY 14.06 FZ 0.04
LOAD 6 WIND LOAD (-X)
JOINT LOAD
3 FX -4.1 FY 2.64 FZ -0.61
7 FX -3.78 FY -3.4 FZ 0.68
60 FX -1.71 FY 2.43 FZ -0.9
61 FX -1.71 FY -1.34 FZ 0.83
1 FX -6.68 FY 7.54 FZ 0
5 FX -6.34 FY -7.87 FZ 0
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
3 FX -1.33 FY 16.22 FZ 6.49
7 FX 0.69 FY 30.35 FZ 10.35
60 FX 0 FY -15.06 FZ 6.98
61 FX 0 FY -27 FZ 11.36
1 FX 0.05 FY -1.16 FZ 2
5 FX 0.6 FY -3.35 FZ 2.01
LOAD 8 WIND LOAD (-Z)
```

```
JOINT LOAD
3 FX 2.69 FY -17.23 FZ -6.62
7 FX -0.47 FY -59.11 FZ -18.25
60 FX 0 FY 15.54 FZ -7.15
61 FX 0 FY 50.88 FZ -20.31
1 FX -0.88 FY 1.69 FZ -2
5 FX -1.34 FY 8.23 FZ -2.05
LOAD 9 TEMPERATURE LOAD 1
JOINT LOAD
3 FX 59.67 FY 3.67 FZ 10.52
7 FX -61.26 FY -3.49 FZ 10.32
60 FX 0 FY -2.77 FZ -8.75
61 FX 0 FY 2.51 FZ -12.12
1 FX 67.93 FY -0.9 FZ 0.01
5 FX -66.34 FY 0.98 FZ 0.02
LOAD 10 TEMPERATURE LOAD 2
JOINT LOAD
3 FX -59.67 FY -3.67 FZ -10.52
7 FX 61.26 FY 3.49 FZ -10.32
60 FX 0 FY 2.77 FZ 8.75
61 FX 0 FY -2.51 FZ 12.12
1 FX -67.93 FY 0.9 FZ -0.01
5 FX 66.34 FY -0.98 FZ -0.02
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
3 FX 1.49 FY -0.21 FZ 0.38
7 FX 1.36 FY 0.42 FZ -0.4
60 FX 0 FY -1.38 FZ 0.49
61 FX 0 FY 1.08 FZ -0.47
1 FX 2.67 FY -2.91 FZ 0
5 FX 2.56 FY 3.01 FZ 0
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
3 FX -0.3 FY 5.63 FZ 1.61
7 FX 0.29 FY 7.23 FZ 2.06
60 FX 0 FY -5.61 FZ 1.93
61 FX 0 FY -7.15 FZ 2.47
1 FX -0.33 FY -0.02 FZ 0
5 FX 0.35 FY -0.08 FZ 0
***LOAD COMBINATION FOR BASE PRESSURE CHECK
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL2 + LL
REPEAT LOAD
```

```
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
```

```
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
```

```
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
```

```
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
```

```
PERFORM ANALYSIS
LOAD LIST 401 TO 438
START CONCRETE DESIGN
CODE ACI 2008
FYMAIN 420000 MEMB 113 TO 116 121 TO 128
FC 35000 MEMB 113 TO 116 121 TO 128
MINMAIN 16 MEMB 113 TO 116 121 TO 133
DESIGN COLUMN 113 TO 116 121 122 124 126 129 TO 132
DESIGN BEAM 123 125 127 128 133 TO 135
CLT 0.05 MEMB 113 TO 116 121 TO 128
END CONCRETE DESIGN
FINISH
```

```
            **************************************************
            *                                                *
            *           STAAD.Pro CONNECT Edition            *
            *           Version  21.00.01.12                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    APR 10, 2020                *
            *           Time=    14:40:40                    *
            *                                                *
            *  Licensed to: MMH KOLKATTA                     *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\TP 4\Civil\Documents\Staad\TP-4 Fdn R0.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-2020
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 5.8 1 4.5; 3 5.8 1 -2; 5 0 1 4.5; 7 0 1 -2; 60 5.8 1 0; 61 0 1 0; 104 0 -2 0
     9. 105 0 -2 4.5; 106 5.8 -2 0; 107 5.8 -2 4.5; 108 0 0.5 0; 109 0 0.5 4.5
    10. 110 5.8 0.5 4.5; 111 5.8 0.5 0; 112 0 -2 -2; 113 0 0.5 -2; 114 5.8 -2 -2
    11. 115 5.8 0.5 -2
    12. MEMBER INCIDENCES
    13. 113 104 108; 114 105 109; 115 107 110; 116 106 111; 121 108 61; 122 109 5
    14. 123 108 109; 124 110 1; 125 109 110; 126 111 60; 127 108 111; 128 111 110
    15. 129 112 113; 130 114 115; 131 113 7; 132 115 3; 133 113 115; 134 113 108
    16. 135 115 111
    17. START GROUP DEFINITION
    18. FLOOR
    19. END GROUP DEFINITION

   ** WARNING: NO VALID GROUPS IN GROUP DEFINITION DATA **
    20. DEFINE MATERIAL START
    21. ISOTROPIC STEEL
    22. E 2.05E+008
    23. POISSON 0.3
    24. DENSITY 76.8195
    25. ALPHA 1.2E-005
    26. DAMP 0.03
    27. TYPE STEEL
    28. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
    29. ISOTROPIC CONCRETE
    30. E 2.17185E+007
    31. POISSON 0.17
    32. DENSITY 23.5616
    33. ALPHA 1E-005
    34. DAMP 0.05
    35. TYPE CONCRETE
    36. STRENGTH FCU 27579
```

```
37. ISOTROPIC ALUMINUM
38. E 6.89476E+007
39. POISSON 0.33
40. DENSITY 26.6018
41. ALPHA 2.3E-005
42. DAMP 0.03
43. END DEFINE MATERIAL
44. MEMBER PROPERTY FRENCH
45. 113 TO 116 121 122 124 126 129 TO 132 PRIS YD 0.6 ZD 0.6
46. 123 125 127 128 133 TO 135 PRIS YD 0.5 ZD 0.3
47. CONSTANTS
48. MATERIAL CONCRETE ALL
49. SUPPORTS
50. 104 TO 107 112 114 FIXED
51. LOAD 1 DEAD  LOAD
52. SELFWEIGHT Y -1
53. JOINT LOAD
54. 3 FX 1.7 FY -16.8 FZ -0.4
55. 7 FX -1.7 FY -25.9 FZ -1.2
56. 60 FX 0 FY -10.8 FZ 1.6
57. 61 FX 0 FY -9.4 FZ 1.2
58. 1 FX 1.5 FY -20.8 FZ -1.3
59. 5 FX -1.5 FY -22.4 FZ 0
60. LOAD 2 LIVE LOAD
61. JOINT LOAD
62. 3 FX 3.2 FY -23.5 FZ -0.3
63. 7 FX -3.1 FY -37.2 FZ -1.7
64. 60 FX 0 FY -31.3 FZ 4.5
65. 61 FX 0 FY -14.2 FZ 1.8
66. 1 FX 3.8 FY -48 FZ -4.3
67. 5 FX -3.9 FY -34.6 FZ 0
68. LOAD 3 SDL-1 EQUIPMENT DEAD LOAD CASE -1.
69. JOINT LOAD
70. 3 FX 22.4 FY -217.5 FZ -44.6
71. 7 FX -10.7 FY -310.5 FZ -69.5
72. 60 FX 0 FY 130.4 FZ -48
73. 61 FX 0 FY 216.1 FZ -77
74. 1 FX 27.1 FY -76.6 FZ 0
75. 5 FX -12.4 FY -82.1 FZ 0
76. LOAD 4 SDL-2 EQUIPMENT DEAD LOAD CASE -2
77. JOINT LOAD
78. 3 FX 17.4 FY -156.2 FZ -27.8
79. 7 FX -8.4 FY -219.6 FZ -43.5
80. 60 FX 0 FY 71.7 FZ -27.9
81. 61 FX 0 FY 125.6 FZ -45.9
82. 1 FX 21 FY -73.9 FZ 0
83. 5 FX -9 FY -81.8 FZ 0
84. LOAD 5 WIND LOAD (+X)
85. JOINT LOAD
86. 3 FX 5.55 FY -3.93 FZ 0.8
87. 7 FX 4.68 FY 11.35 FZ -1.31
88. 60 FX 1.71 FY -3.21 FZ 1.2
89. 61 FX 1.69 FY -7.51 FZ -0.73
90. 1 FX 9.32 FY -10.76 FZ 0
91. 5 FX 7.75 FY 14.06 FZ 0.04
92. LOAD 6 WIND LOAD (-X)
```

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 93. JOINT LOAD
 94. 3 FX -4.1 FY 2.64 FZ -0.61
 95. 7 FX -3.78 FY -3.4 FZ 0.68
 96. 60 FX -1.71 FY 2.43 FZ -0.9
 97. 61 FX -1.71 FY -1.34 FZ 0.83
 98. 1 FX -6.68 FY 7.54 FZ 0
 99. 5 FX -6.34 FY -7.87 FZ 0
100. LOAD 7 WIND LOAD (+Z)
101. JOINT LOAD
102. 3 FX -1.33 FY 16.22 FZ 6.49
103. 7 FX 0.69 FY 30.35 FZ 10.35
104. 60 FX 0 FY -15.06 FZ 6.98
105. 61 FX 0 FY -27 FZ 11.36
106. 1 FX 0.05 FY -1.16 FZ 2
107. 5 FX 0.6 FY -3.35 FZ 2.01
108. LOAD 8 WIND LOAD (-Z)
109. JOINT LOAD
110. 3 FX 2.69 FY -17.23 FZ -6.62
111. 7 FX -0.47 FY -59.11 FZ -18.25
112. 60 FX 0 FY 15.54 FZ -7.15
113. 61 FX 0 FY 50.88 FZ -20.31
114. 1 FX -0.88 FY 1.69 FZ -2
115. 5 FX -1.34 FY 8.23 FZ -2.05
116. LOAD 9 TEMPERATURE LOAD 1
117. JOINT LOAD
118. 3 FX 59.67 FY 3.67 FZ 10.52
119. 7 FX -61.26 FY -3.49 FZ 10.32
120. 60 FX 0 FY -2.77 FZ -8.75
121. 61 FX 0 FY 2.51 FZ -12.12
122. 1 FX 67.93 FY -0.9 FZ 0.01
123. 5 FX -66.34 FY 0.98 FZ 0.02
124. LOAD 10 TEMPERATURE LOAD 2
125. JOINT LOAD
126. 3 FX -59.67 FY -3.67 FZ -10.52
127. 7 FX 61.26 FY 3.49 FZ -10.32
128. 60 FX 0 FY 2.77 FZ 8.75
129. 61 FX 0 FY -2.51 FZ 12.12
130. 1 FX -67.93 FY 0.9 FZ -0.01
131. 5 FX 66.34 FY -0.98 FZ -0.02
132. LOAD 11 SEISMIC LOAD (+X)
133. JOINT LOAD
134. 3 FX 1.49 FY -0.21 FZ 0.38
135. 7 FX 1.36 FY 0.42 FZ -0.4
136. 60 FX 0 FY -1.38 FZ 0.49
137. 61 FX 0 FY 1.08 FZ -0.47
138. 1 FX 2.67 FY -2.91 FZ 0
139. 5 FX 2.56 FY 3.01 FZ 0
140. LOAD 12 SEISMIC LOAD (+Z)
141. JOINT LOAD
142. 3 FX -0.3 FY 5.63 FZ 1.61
143. 7 FX 0.29 FY 7.23 FZ 2.06
144. 60 FX 0 FY -5.61 FZ 1.93
145. 61 FX 0 FY -7.15 FZ 2.47
146. 1 FX -0.33 FY -0.02 FZ 0
147. 5 FX 0.35 FY -0.08 FZ 0
148. ***LOAD COMBINATION FOR BASE PRESSURE CHECK
```

```
149. LOAD 301 DL+SDL1
150. REPEAT LOAD
151. 1 1.0 3 1.0
152. LOAD 302 DL+SDL2
153. REPEAT LOAD
154. 1 1.0 4 1.0
155. LOAD 303 DL + SDL1 + TL1 + LL
156. REPEAT LOAD
157. 1 1.0 3 1.0 9 1.0 2 1.0
158. LOAD 304 DL + SDL1 + TL1 + LL
159. REPEAT LOAD
160. 1 1.0 4 1.0 9 1.0 2 1.0
161. LOAD 305 DL + SDL2 + TL2 + LL
162. REPEAT LOAD
163. 1 1.0 3 1.0 10 1.0 2 1.0
164. LOAD 306 DL + SDL2 + TL2 + LL
165. REPEAT LOAD
166. 1 1.0 4 1.0 10 1.0 2 1.0
167. LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
168. REPEAT LOAD
169. 1 1.0 3 1.0 2 0.75 5 0.75
170. LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
171. REPEAT LOAD
172. 1 1.0 3 1.0 2 0.75 6 0.75
173. LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
174. REPEAT LOAD
175. 1 1.0 3 1.0 2 0.75 7 0.75
176. LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
177. REPEAT LOAD
178. 1 1.0 3 1.0 2 0.75 8 0.75
179. LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
180. REPEAT LOAD
181. 1 1.0 4 1.0 2 0.75 5 0.75
182. LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
183. REPEAT LOAD
184. 1 1.0 4 1.0 2 0.75 6 0.75
185. LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
186. REPEAT LOAD
187. 1 1.0 4 1.0 2 0.75 7 0.75
188. LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
189. REPEAT LOAD
190. 1 1.0 4 1.0 2 0.75 8 0.75
191. LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
192. REPEAT LOAD
193. 1 1.0 3 1.0 2 0.75 11 0.525
194. LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
195. REPEAT LOAD
196. 1 1.0 3 1.0 2 0.75 11 -0.525
197. LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
198. REPEAT LOAD
199. 1 1.0 3 1.0 2 0.75 12 0.525
200. LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
201. REPEAT LOAD
202. 1 1.0 3 1.0 2 0.75 12 -0.525
203. LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
204. REPEAT LOAD
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205. 1 1.0 4 1.0 2 0.75 11 0.525
206. LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
207. REPEAT LOAD
208. 1 1.0 4 1.0 2 0.75 11 -0.525
209. LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
210. REPEAT LOAD
211. 1 1.0 4 1.0 2 0.75 12 0.525
212. LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
213. REPEAT LOAD
214. 1 1.0 4 1.0 2 0.75 12 -0.525
215. LOAD 323 0.6 (DL + SDL1) + WL(+X)
216. REPEAT LOAD
217. 1 0.6 3 0.6 5 1.0
218. LOAD 324 0.6 (DL + SDL1) + WL(-X)
219. REPEAT LOAD
220. 1 0.6 3 0.6 6 1.0
221. LOAD 325 0.6 (DL + SDL1) + WL(+Z)
222. REPEAT LOAD
223. 1 0.6 3 0.6 7 1.0
224. LOAD 326 0.6 (DL + SDL1) + WL(-Z)
225. REPEAT LOAD
226. 1 0.6 3 0.6 8 1.0
227. LOAD 327 0.6 (DL + SDL2) + WL(+X)
228. REPEAT LOAD
229. 1 0.6 4 0.6 5 1.0
230. LOAD 328 0.6 (DL + SDL2) + WL(-X)
231. REPEAT LOAD
232. 1 0.6 4 0.6 6 1.0
233. LOAD 329 0.6 (DL + SDL2) + WL(+Z)
234. REPEAT LOAD
235. 1 0.6 4 0.6 7 1.0
236. LOAD 330 0.6 (DL + SDL2) + WL(-Z)
237. REPEAT LOAD
238. 1 0.6 4 0.6 8 1.0
239. LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
240. REPEAT LOAD
241. 1 0.6 3 0.6 11 0.7
242. LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
243. REPEAT LOAD
244. 1 0.6 3 0.6 11 -0.7
245. LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
246. REPEAT LOAD
247. 1 0.6 3 0.6 12 0.7
248. LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
249. REPEAT LOAD
250. 1 0.6 3 0.6 12 -0.7
251. LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
252. REPEAT LOAD
253. 1 0.6 4 0.6 11 0.7
254. LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
255. REPEAT LOAD
256. 1 0.6 4 0.6 11 -0.7
257. LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
258. REPEAT LOAD
259. 1 0.6 4 0.6 12 0.7
260. LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
```

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261. REPEAT LOAD
262. 1 0.6 4 0.6 12 -0.7
263. ****
264. *LOAD COMBINATION FOR CONCRETE DESIGN
265. LOAD 401 1.4 (DL+SDL1)
266. REPEAT LOAD
267. 1 1.4 3 1.4
268. LOAD 402 1.4 (DL+SDL2)
269. REPEAT LOAD
270. 1 1.4 4 1.4
271. LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
272. REPEAT LOAD
273. 1 1.4 3 1.4 9 1.4 2 1.7
274. LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
275. REPEAT LOAD
276. 1 1.4 4 1.4 9 1.4 2 1.7
277. LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
278. REPEAT LOAD
279. 1 1.4 3 1.4 10 1.4 2 1.7
280. LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
281. REPEAT LOAD
282. 1 1.4 4 1.4 10 1.4 2 1.7
283. LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
284. REPEAT LOAD
285. 1 1.2 3 1.2 2 0.5 5 1.6
286. LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
287. REPEAT LOAD
288. 1 1.2 3 1.2 2 0.5 6 1.6
289. LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
290. REPEAT LOAD
291. 1 1.2 3 1.2 2 0.5 7 1.6
292. LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
293. REPEAT LOAD
294. 1 1.2 3 1.2 2 0.5 8 1.6
295. LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
296. REPEAT LOAD
297. 1 1.2 4 1.2 2 0.5 5 1.6
298. LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
299. REPEAT LOAD
300. 1 1.2 4 1.2 2 0.5 6 1.6
301. LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
302. REPEAT LOAD
303. 1 1.2 4 1.2 2 0.5 7 1.6
304. LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
305. REPEAT LOAD
306. 1 1.2 4 1.2 2 0.5 8 1.6
307. LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
308. REPEAT LOAD
309. 1 1.2 3 1.2 2 0.5 11 1.0
310. LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
311. REPEAT LOAD
312. 1 1.2 3 1.2 2 0.5 11 -1.0
313. LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
314. REPEAT LOAD
315. 1 1.2 3 1.2 2 0.5 12 1.0
316. LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
```

```
317. REPEAT LOAD
318. 1 1.2 3 1.2 2 0.5 12 -1.0
319. LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
320. REPEAT LOAD
321. 1 1.2 4 1.2 2 0.5 11 1.0
322. LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
323. REPEAT LOAD
324. 1 1.2 4 1.2 2 0.5 11 -1.0
325. LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
326. REPEAT LOAD
327. 1 1.2 4 1.2 2 0.5 12 1.0
328. LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
329. REPEAT LOAD
330. 1 1.2 4 1.2 2 0.5 12 -1.0
331. LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
332. REPEAT LOAD
333. 1 0.9 3 0.9 5 1.6
334. LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
335. REPEAT LOAD
336. 1 0.9 3 0.9 6 1.6
337. LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
338. REPEAT LOAD
339. 1 0.9 3 0.9 7 1.6
340. LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
341. REPEAT LOAD
342. 1 0.9 3 0.9 8 1.6
343. LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
344. REPEAT LOAD
345. 1 0.9 4 0.9 5 1.6
346. LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
347. REPEAT LOAD
348. 1 0.9 4 0.9 6 1.6
349. LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
350. REPEAT LOAD
351. 1 0.9 4 0.9 7 1.6
352. LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
353. REPEAT LOAD
354. 1 0.9 4 0.9 8 1.6
355. LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
356. REPEAT LOAD
357. 1 0.9 3 0.9 11 1.0
358. LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
359. REPEAT LOAD
360. 1 0.9 3 0.9 11 -1.0
361. LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
362. REPEAT LOAD
363. 1 0.9 3 0.9 12 1.0
364. LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
365. REPEAT LOAD
366. 1 0.9 3 0.9 12 -1.0
367. LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
368. REPEAT LOAD
369. 1 0.9 4 0.9 11 1.0
370. LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
371. REPEAT LOAD
372. 1 0.9 4 0.9 11 -1.0
```

```
373. LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
374. REPEAT LOAD
375. 1 0.9 4 0.9 12 1.0
376. LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
377. REPEAT LOAD
378. 1 0.9 4 0.9 12 -1.0
379. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 88, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
380. LOAD LIST 401 TO 438
381. START CONCRETE DESIGN
382. CODE ACI 2008
383. FYMAIN 420000 MEMB 113 TO 116 121 TO 128
384. FC 35000 MEMB 113 TO 116 121 TO 128
385. MINMAIN 16 MEMB 113 TO 116 121 TO 133
386. DESIGN COLUMN 113 TO 116 121 122 124 126 129 TO 132
```

==========

COLUMN NO. 113 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 114 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 115 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 116 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 121 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 122 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 124 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 126 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 129 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 130 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 131 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

---

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 132 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

---

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

**********END OF COLUMN DESIGN RESULTS**********

387. DESIGN BEAM 123 125 127 128 133 TO 135

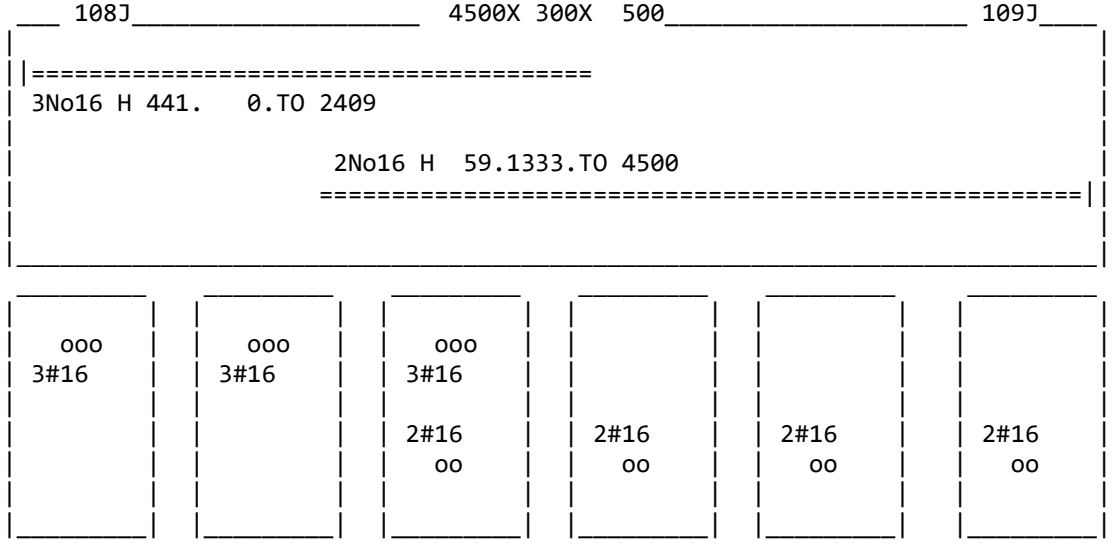
BEAM NO. 123 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	1333.	4500.	NO	YES
2	441.	3 - 16MM	0.	2409.	YES	NO

B E A M N O. 123 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.06 KNS Vc= 114.16 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.11 KNS Vc= 114.16 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



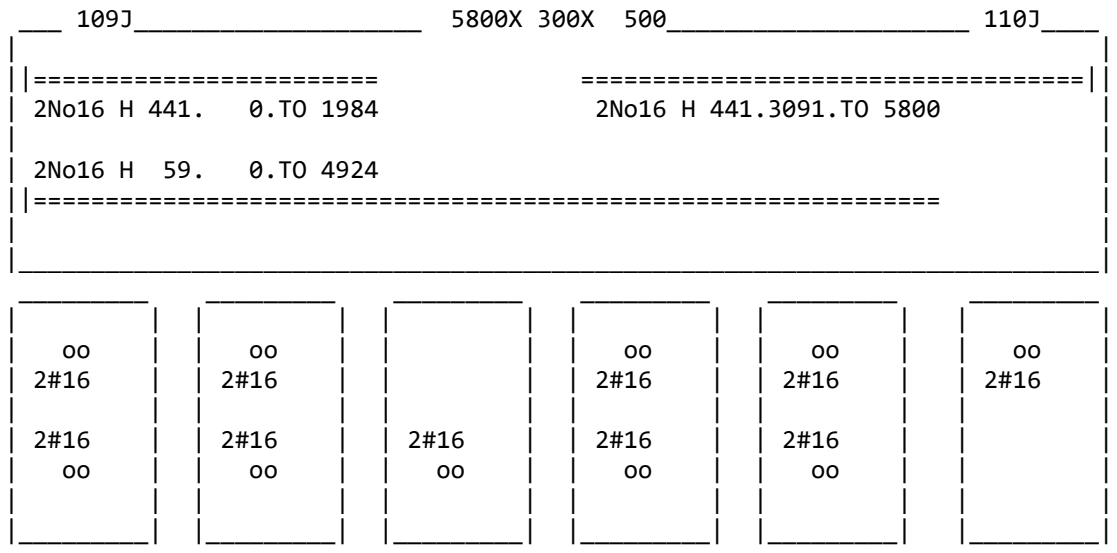
BEAM NO. 125 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5800. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	4924.	YES	NO
2	441.	2 - 16MM	0.	1984.	YES	NO
3	441.	2 - 16MM	3091.	5800.	NO	YES

B E A M N O. 125 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.74 KNS Vc= 123.01 KNS Vs= 0.00 KNS
Tu= 0.67 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.64 KNS Vc= 123.01 KNS Vs= 0.00 KNS
Tu= 0.67 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



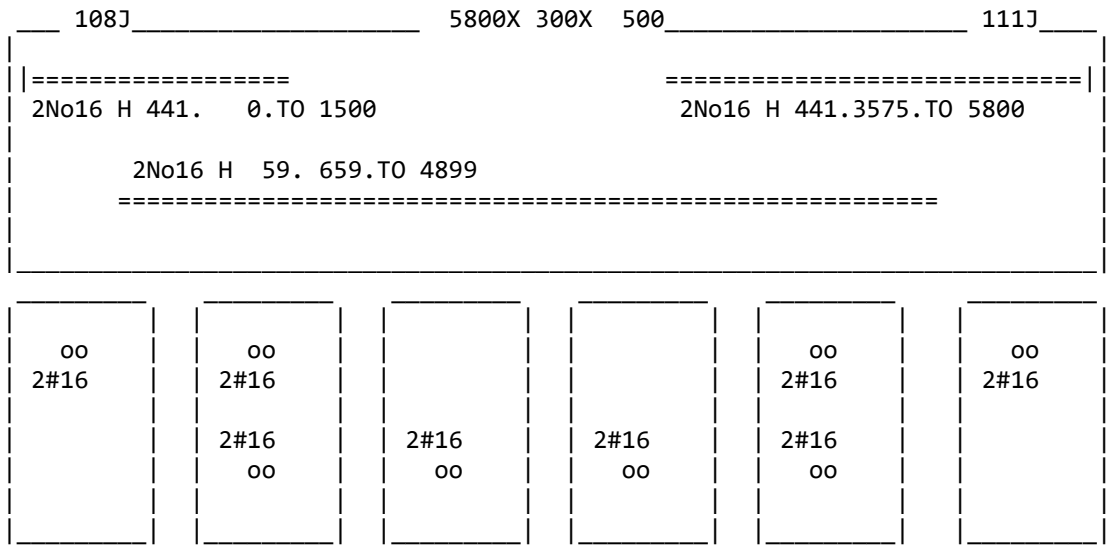
BEAM NO. 127 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5800. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	659.	4899.	NO	NO
2	441.	2 - 16MM	0.	1500.	YES	NO
3	441.	2 - 16MM	3575.	5800.	NO	YES

B E A M N O. 127 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.78 KNS Vc= 124.89 KNS Vs= 0.00 KNS
Tu= 0.75 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.60 KNS Vc= 124.62 KNS Vs= 0.00 KNS
Tu= 0.75 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



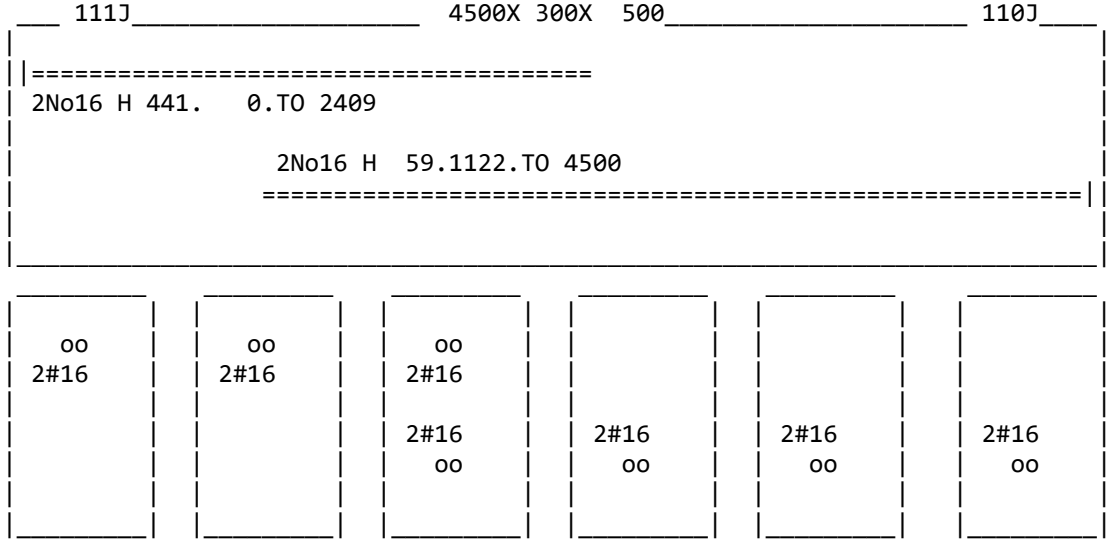
BEAM NO. 128 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	1122.	4500.	NO	YES
2	441.	2 - 16MM	0.	2409.	YES	NO

B E A M N O. 128 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.56 KNS Vc= 121.02 KNS Vs= 0.00 KNS
Tu= 0.77 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.62 KNS Vc= 121.02 KNS Vs= 0.00 KNS
Tu= 0.77 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



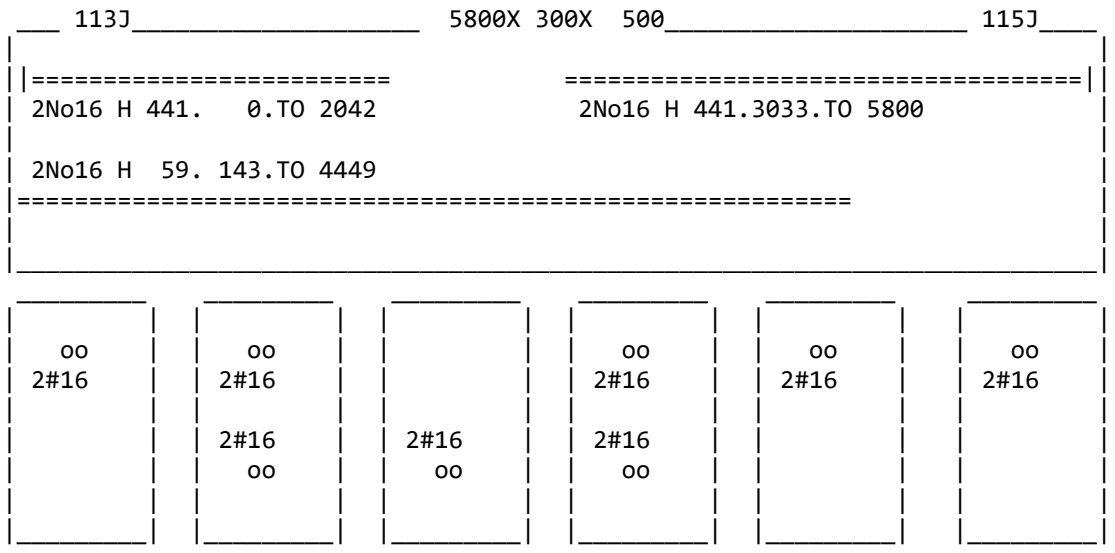
BEAM NO. 133 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5800. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	143.	4449.	NO	NO
2	441.	2 - 16MM	0.	2042.	YES	NO
3	441.	2 - 16MM	3033.	5800.	NO	YES

B E A M N O. 133 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.31 KNS Vc= 110.46 KNS Vs= 0.00 KNS
Tu= 0.90 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.07 KNS Vc= 110.46 KNS Vs= 0.00 KNS
Tu= 0.90 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 134 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

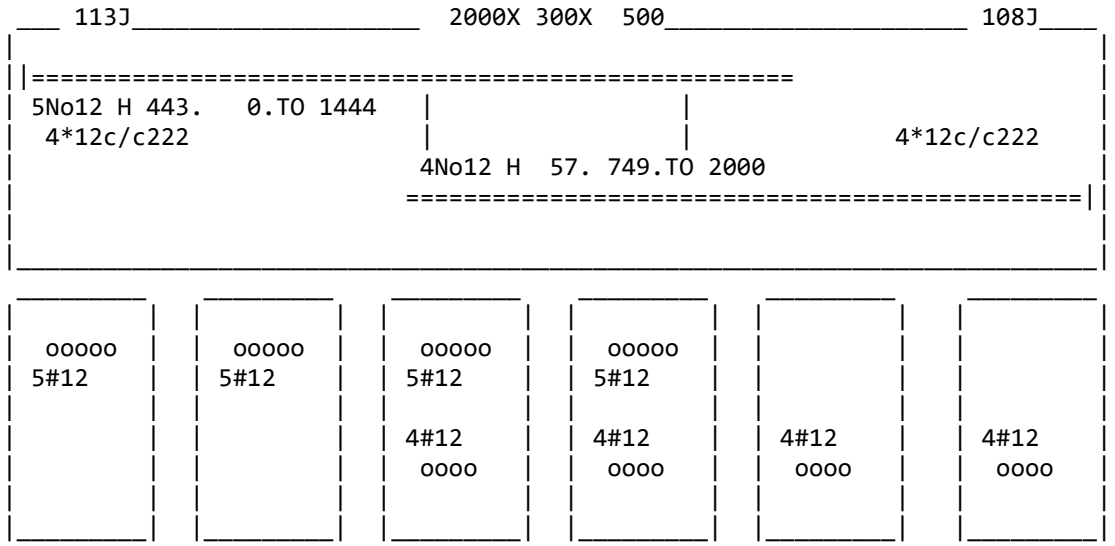
LEN - 2000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	749.	2000.	NO	YES
2	443.	5 - 12MM	0.	1444.	YES	NO

B E A M N O. 134 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.64 KNS Vc= 107.55 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.2 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 564. MM

AT END SUPPORT - Vu= 62.06 KNS Vc= 107.55 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.2 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 564. MM



BEAM NO. 135 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

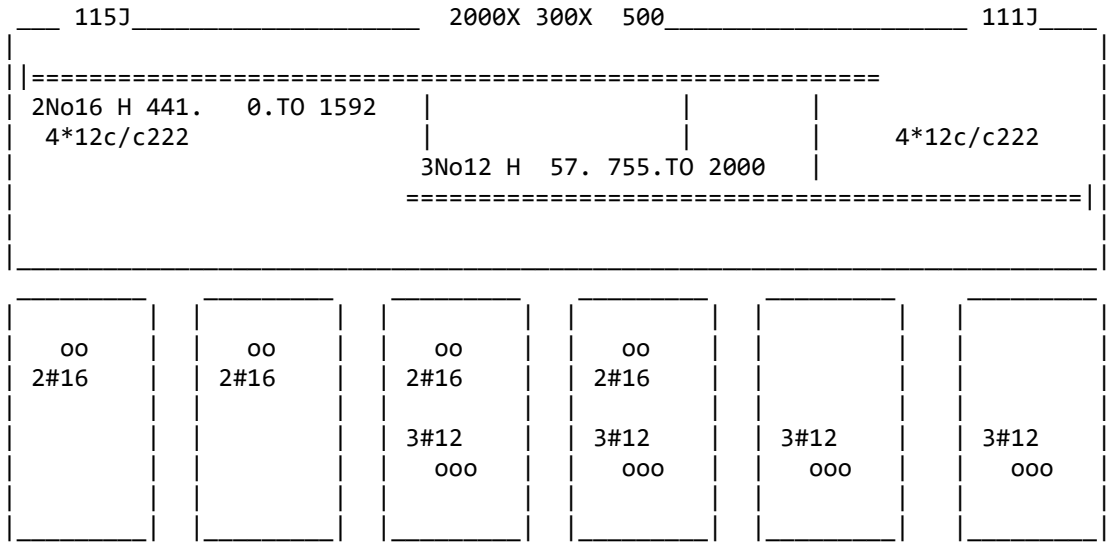
LEN - 2000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	755.	2000.	NO	YES
2	441.	2 - 16MM	0.	1592.	YES	NO

B E A M N O. 135 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 44.28 KNS Vc= 110.63 KNS Vs= 0.00 KNS
Tu= 1.19 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 401
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 564. MM

AT END SUPPORT - Vu= 41.92 KNS Vc= 110.73 KNS Vs= 0.00 KNS
Tu= 1.27 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 410
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 564. MM



**********END OF BEAM DESIGN**********

388. CLT 0.05 MEMB 113 TO 116 121 TO 128
389. END CONCRETE DESIGN
390. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 10,2020 TIME= 14:40:45 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 4 STAAD INPUT FILE OF FOOTING F2

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 9-Apr-2020
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
104 0 -2.9 2; 112 0 -2.9 0; 200 0 -3 0; 201 0 -3 2; 204 0 -3 -1.75;
205 0 -3 3.75; 206 1.75 -3 -1.75; 207 1.75 -3 3.75; 208 -1.75 -3 -1.75;
209 -1.75 -3 3.75; 210 -0.75 -3 3.75; 211 -0.75 -3 2.75; 212 -1.75 -3 2.75;
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394 -0.25 -3 -1.475; 395 0 -3 -1.475; 396 0.25 -3 -1.475; 397 0.5 -3 -1.475;
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454 1.31667 -3 3.475; 455 -1.31667 -3 2.75; 456 -1.31667 -3 2.5;
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MEMBER INCIDENCES
1 200 112; 3 201 104;
ELEMENT INCIDENCES SHELL
15 211 214 227 225; 16 214 216 228 227; 17 216 217 229 228; 18 217 219 230 229;
19 219 221 231 230; 20 221 223 232 231; 23 225 227 236 234; 24 227 228 237 236;
25 228 229 238 237; 26 229 230 239 238; 27 230 231 240 239; 28 231 232 241 240;
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35 239 240 248 247; 36 240 241 249 248; 39 243 245 253 251; 40 245 246 254 253;
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67 274 275 284 283; 68 275 276 285 284; 71 278 280 289 287; 72 280 281 290 289;
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79 287 289 298 296; 80 289 290 299 298; 81 290 291 300 299; 82 291 292 301 300;
83 292 293 302 301; 84 293 294 303 302; 87 296 298 307 305; 88 298 299 308 307;
89 299 300 309 308; 90 300 301 310 309; 91 301 302 311 310; 92 302 303 312 311;
95 305 307 316 314; 96 307 308 317 316; 97 308 309 200 317; 98 309 310 318 200;
99 310 311 319 318; 100 311 312 320 319; 103 314 316 324 322;
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107 318 319 328 327; 108 319 320 329 328; 111 322 324 333 331;
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147 363 365 366 364; 149 345 346 367 365; 150 365 367 368 366;
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172 382 384 219 217; 174 383 385 386 384; 175 384 386 221 219;
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188 362 364 395 394; 189 394 395 204 351; 190 364 366 396 395;
191 395 396 352 204; 192 366 368 397 396; 193 396 397 353 352;
194 368 370 398 397; 195 397 398 354 353; 200 210 213 402 400;
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210 220 222 407 406; 211 406 407 387 385; 214 223 409 410 232;
215 409 411 412 410; 216 411 224 233 412; 217 232 410 413 241;
218 410 412 414 413; 219 412 233 242 414; 220 241 413 415 249;
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224 415 416 418 417; 225 416 250 259 418; 226 258 417 419 267;
227 417 418 420 419; 228 418 259 268 420; 229 267 419 421 276;
230 419 420 422 421; 231 420 268 277 422; 232 276 421 423 285;
233 421 422 424 423; 234 422 277 286 424; 235 285 423 425 294;
236 423 424 426 425; 237 424 286 295 426; 238 294 425 427 303;
239 425 426 428 427; 240 426 295 304 428; 241 303 427 429 312;
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248 431 432 434 433; 249 432 321 330 434; 250 329 433 435 338;
251 433 434 436 435; 252 434 330 339 436; 253 338 435 437 347;
254 435 436 438 437; 255 436 339 348 438; 256 347 437 439 369;
257 437 438 440 439; 258 438 348 371 440; 259 369 439 441 370;
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266 444 446 411 409; 267 446 390 224 411; 268 370 441 447 398;
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275 451 453 454 452; 276 453 207 408 454; 277 407 452 443 387;
278 452 454 445 443; 279 454 408 389 445; 280 212 455 456 226;
281 455 457 458 456; 282 457 211 225 458; 283 226 456 459 235;
284 456 458 460 459; 285 458 225 234 460; 286 235 459 461 244;
287 459 460 462 461; 288 460 234 243 462; 289 244 461 463 252;
290 461 462 464 463; 291 462 243 251 464; 292 252 463 465 261;
293 463 464 466 465; 294 464 251 260 466; 295 261 465 467 270;
296 465 466 468 467; 297 466 260 269 468; 298 270 467 469 279;
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302 469 470 472 471; 303 470 278 287 472; 304 288 471 473 297;
305 471 472 474 473; 306 472 287 296 474; 307 297 473 475 306;
308 473 474 476 475; 309 474 296 305 476; 310 306 475 477 315;
311 475 476 478 477; 312 476 305 314 478; 313 315 477 479 323;

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314 477 478 480 479; 315 478 314 322 480; 316 323 479 481 332;
317 479 480 482 481; 318 480 322 331 482; 319 332 481 483 341;
320 481 482 484 483; 321 482 331 340 484; 322 341 483 485 356;
323 483 484 486 485; 324 484 340 355 486; 325 356 485 487 358;
326 485 486 488 487; 327 486 355 357 488; 328 374 489 490 376;
329 489 491 492 490; 330 491 373 375 492; 331 376 490 455 212;
332 490 492 457 455; 333 492 375 211 457; 334 358 487 493 392;
335 487 488 494 493; 336 488 357 391 494; 337 392 493 495 208;
338 493 494 496 495; 339 494 391 349 496; 340 209 497 498 401;
341 497 499 500 498; 342 499 210 400 500; 343 401 498 489 374;
344 498 500 491 489; 345 500 400 373 491;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 3 PRIS YD 0.7 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
200 201 204 TO 500 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
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143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -29
JOINT LOAD
104 FX -5.1 FY -57.7 FZ -2.5
104 MX -2.1 MY 0 MZ 3.6
112 FX -4.8 FY -64.2 FZ -0.2
112 MX -0.4 MY 0 MZ 3.5
LOAD 2 LIVE LOAD
JOINT LOAD
104 FX 0.1 FY -14.6 FZ -0.3
104 MX -0.3 MY 0 MZ -0.1
112 FX 0.7 FY -36.8 FZ 0.5
112 MX 0.2 MY 0.1 MZ -0.3
LOAD 3 SDL-1 EQUIPMENT DEAD LOAD 1
JOINT LOAD
104 FX 1.7 FY 250.2 FZ -54.8
104 MX -99.6 MY -0.7 MZ -3.6
112 FX 8.3 FY -356.4 FZ -43
112 MX -91.8 MY -1.2 MZ -12.1
LOAD 4 SDL-2 EQUIPMENT DEAD LOAD 2
JOINT LOAD
104 FX 1.4 FY 146.1 FZ -33.6
104 MX -61 MY -0.5 MZ -2.9
112 FX 6.3 FY -247.1 FZ -25.9
112 MX -55.9 MY -0.8 MZ -9.3
LOAD 5 WIND LOAD (+X)
JOINT LOAD
104 FX 2.79 FY -5.82 FZ -0.83
104 MX -1.3 MY -0.1 MZ -6.3
112 FX 4.44 FY 11.43 FZ -0.66
112 MX -1.2 MY -0.6 MZ -9.98
LOAD 6 WIND LOAD (-X)
JOINT LOAD
104 FX -2.43 FY -2.45 FZ 0.55
104 MX 1 MY 0.1 MZ 5.51
112 FX -3.48 FY -3.7 FZ 0.49
112 MX 0.9 MY 0.4 MZ 7.87
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
104 FX -0.03 FY -32.73 FZ 9.12
104 MX 16.2 MY 0.1 MZ 0.08
112 FX -0.51 FY 38.13 FZ 7.22
112 MX 14.9 MY 0.1 MZ 0.72
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
104 FX 0.11 FY 60.6 FZ -15.35
104 MX -27.5 MY -0.4 MZ -0.29
112 FX 1.35 FY -72.16 FZ -12.2
112 MX -25.4 MY -0.4 MZ -2.3
LOAD 9 TEMPERATURE LOAD 1
JOINT LOAD
104 FX 0.86 FY 2.31 FZ 1.74
104 MX 0.6 MY 0.8 MZ -0.29
112 FX 11.74 FY -3.86 FZ -2.64
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112 MX -2.2 MY 0 MZ -3.31
LOAD 10 TEMPERATURE LOAD 2
JOINT LOAD
104 FX -0.86 FY -2.31 FZ -1.74
104 MX -0.6 MY -0.77 MZ 0.29
112 FX -11.74 FY 3.86 FZ 2.64
112 MX 2.15 MY -1.16 MZ 3.31
LOAD 11 EARTHQUAKE LOAD (+X)
JOINT LOAD
104 FX 0.44 FY 1.44 FZ -0.32
104 MX -0.6 MY -0.1 MZ -1
112 FX 1.12 FY 0.4 FZ -0.27
112 MX -0.5 MY 0 MZ -2.55
LOAD 12 EARTHQUAKE LOAD (+Z)
JOINT LOAD
104 FX 0 FY -8.23 FZ 1.68
104 MX 3.1 MY 0 MZ 0
112 FX -0.07 FY 8.72 FZ 1.36
112 MX 2.9 MY 0 MZ 0.04
*****************************************
***LOAD COMBINATION FOR BASE PRESSURE CHECK
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
```

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LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
```

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LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
```

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REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
```

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REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 3
FYMAIN 420000 MEMB 1 3
TRACK 1 ALL
CLB 0.05 MEMB 1 3
CLT 0.05 MEMB 1 3
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 ALL
DESIGN COLUMN 1 3
END CONCRETE DESIGN
FINISH
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	10-04-2020	11-05-2020	01-11-2023				Date	10-04-2020	11-05-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT DATED 05-05-20	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 5 **COMMENT RESOLUTION SHEET**

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002343, 5 May 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 5 May 2020	Comment by	EPC Response DATE: 11 May 2020	Status	Remarks
			General Comments (cc)				
MD-216-3130-EG-CV-CAL-1006-A01	3130	1	see comment on Design of TP-1	Mr. Doron Golan Bechtel	The description has been changed as commented in TP1	Closed	